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REGULAR PAPER

Efficient and cost-effective influence and blocker minimization

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Abstract

Online social networks have emerged as prominent platforms for individuals to rapidly share ideas and perspectives. However, their swift dissemination capabilities also make them powerful channels for the spread of misinformation. Such dissemination results in substantial economic and societal harm, highlighting the need for effective suppression. To support this practical demand, this paper investigates the influence minimization (IMIN) and blocker minimization (BM) problems. The IMIN problem aims to find a node set of size k (the given budget) to block, such that the reduction in influence spread of a given seed set S is maximized. The BM problem is the dual of IMIN, aiming to find a node set of minimum cardinality to block, such that the threshold condition is satisfied, i.e., the reduction in influence spread of S is not less than the given threshold η. Both problems are NP-hard and involve a non-submodular objective function. Existing IMIN solutions incur high computational costs and offer no approximation guarantees on influence spread reduction. To fill the gap, we build upon the sandwich strategy to develop the first efficient algorithm with data-dependent approximation guarantees for IMIN. In particular, we design non-trivial $(1-1/e-\epsilon)$-approximation algorithms to optimize the proposed submodular bounding functions. For the BM problem, the existing solution is of limited effectiveness and lacks theoretical guarantees for both node set cardinality and threshold condition satisfaction. In this paper, we design a heuristic and two bicriteria approximation algorithms to address the BM problem. The heuristic is efficient and satisfies the threshold condition, but fails to ensure near-optimal cardinality. The two approximation algorithms both provide guarantees on cardinality: the first attains a tighter approximation ratio at the expense of efficiency, while the second yields a slightly weaker ratio with improved efficiency. Finally, comprehensive experiments on 9 real-world datasets are conducted to validate the efficiency and effectiveness of the proposed techniques.

Keywords Social networks · Influence minimization · Non-submodular optimization

1 Introduction

With the rapid development of the Internet, various online social networks (OSNs) have thrived, immensely satisfying and facilitating the need for individuals to share their perspectives. Leveraging the established connections among individuals, ideas spread across OSNs through word-of-mouth effects [12]. However, immense user bases and rapid sharing abilities also make OSNs effective channels for spreading misinformation, which leads to significant harm, such as economic damages and societal unrest [3, 36, 59]. As a result, limiting the spread of misinformation has emerged as an important and urgent task. To this end, one effective strategy is to remove a set of critical nodes (referred to as *blockers*) from the network. A natural question under this strategy is, within a limited budget, how to identify a set of blockers that maximally suppresses misinformation. Wang et al. [52] first formulate this as an optimization problem, referred to as

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influence minimization (IMIN). Formally, given a graph G, a seed set S, and a budget k, the IMIN problem aims to find a set of k nodes to block such that the reduction in influence spread of S is maximized. In addition, in real-world information diffusion, user engagement may reach a tipping point beyond which the diffusion process unfolds like wildfire [15, 35]. Hence, those seeking to suppress misinformation would like to prevent a target number of users from being exposed, thereby avoiding the tipping point and curbing its explosive spread. To support this practical requirement, Pham et al. [41] formulate the *blocker minimization* (BM) problem: given a graph G, a seed set S, and a threshold η, the BM problem aims to find a blocker set with minimum cardinality such that the *threshold condition* is satisfied, i.e., the reduction in influence spread of S is not less than the given threshold η. In this paper, we investigate both the IMIN and BM problems under the IC model, and aim to develop algorithms that are practically efficient and theoretically grounded.

The IMIN problem is NP-hard and APX-hard under the IC model [60]. To address the problem, Wang et al. [52] first utilize Monte-Carlo simulations to estimate the reduced influence spread of seed set S given any blocker, and provide a greedy algorithm to select blockers, i.e., iteratively select the node that leads to the largest reduced influence spread. However, their solution is prohibitively expensive on large graphs due to the inefficiency of Monte-Carlo simulations. Recently, based on the dominator tree (see Sect. 2.4 for formal definition), Xie et al. [60] design an efficient approach that can accurately estimate the reduced spread of S. On this basis, a greedy method is first proposed. Xie et al. further observe that the greedy framework may miss selecting some critical nodes. This motivates the development of a more effective modified greedy algorithm, which prioritizes the outgoing neighbors of seed nodes in S. However, the solutions in [60] require re-estimating the reduced spread of S after selecting a blocker in each iteration, which results in prohibitive computation overhead, especially when the budget k becomes large. Moreover, since the objective function of IMIN is non-submodular, adopting the greedy method cannot provide $(1-1/e)$-approximate solutions [37]. Based on the notions of curvature and submodularity ratio [4], we further prove that applying the greedy method to the IMIN objective cannot provide any meaningful performance guarantees on influence spread reduction.

To bridge the gap in IMIN, we propose SandIMIN, an approximation algorithm which exhibits superior efficiency and provides strong approximation guarantees. Building on the sandwich strategy [34, 56], SandIMIN first develops submodular lower and upper bounds for the objective function, and then produces solutions with $(1-1/e-\epsilon)$-approximations to maximize the lower and upper bounding functions (breads of the sandwich). A heuristic solution for IMIN is also produced (filling of the sandwich). SandIMIN finally returns the best result among the three candidate solutions.

The development of SandIMIN is non-trivial due to the three challenges. Firstly, tight bounding functions with the submodularity property (w.r.t. the IMIN objective) are required to provide high-quality results. In particular, the empirical effectiveness and approximation guarantees of the sandwich-based approach rely on how close the proposed bounds are to the objective function; a loose bounding function can be applied in our scenario, but it would fail to produce satisfactory results. For example, a constant function can serve as a trivial submodular upper bound (e.g., an upper bound that equals the number of nodes in the graph). However, it is apparent that this bounding function may adversely affect our results since the solution to the constant function can be arbitrary. We address this challenge by developing robust lower and upper bounding functions, with their tightness empirically evidenced by SandIMIN's superior effectiveness and approximation ratio in our experiments.

Secondly, the proposed bounding functions are intractable to compute in practice, and we prove that their computations are both #P-hard. This necessitates efficient and accurate estimation methods. The widely used and powerful technique for influence estimation, reverse influence sampling (RIS) [5], cannot be trivially adapted to our setting. This is because, in contrast with RIS that treats all the nodes equally and samples the node uniformly, we only need to focus on the nodes that are prone to be activated by the seed set S, since only those nodes can contribute to the bounding functions. That is, if we estimate the bounding functions using a similar manner of RIS, we need to sample the node based on its probability of being activated by S. However, this probability is #P-hard to compute [7]. To overcome this issue, we first define two new sample sets named CP sequence and LRR set, and design efficient approaches for their generation. On this basis, the unbiased estimators for the bounding functions are then established.

Thirdly, it is essential to derive the appropriate sample complexity for bounding function maximization to ensure both performance guarantees and practical implementation. The sampling schemes used in existing influence maximization solutions (e.g., OPIM-C [43]) cannot be applied to address this challenge, as their unbiased estimator for influence spread differs in form from ours. In view of this, we first establish novel martingale-based concentration bounds tailored to our unbiased estimators. With these results, we develop advanced stopping conditions to correctly mitigate the redundant sample generation. Building on these foundations, we design App-LB and App-UB, $(1-1/e-\epsilon)$-approximation algorithms for lower and upper bounding functions maximization, respectively.

The BM problem is also proven to be NP-hard under the IC model [41]. To tackle the problem, Pham et al. [41] propose

a greedy approach that iteratively selects the blocker with the largest estimated marginal gain in influence spread reduction, until the threshold condition is satisfied. In particular, they utilize a tree-based estimation method for the objective function, which is efficient but inherently yields biased results. As a consequence, the resulting algorithm exhibits limited effectiveness and fails to provide any theoretical guarantees for both blocker set cardinality and threshold condition satisfaction.

In this work, we propose three more effective algorithms for the BM problem. Our first method, Heur-BM, follows the general workflow of the approach proposed in [41] but improves it by incorporating an unbiased estimator of the objective function. This design enables Heur-BM to theoretically satisfy the threshold condition with high probability. In addition, Heur-BM attains high efficiency through a novel two-stage algorithmic optimization. Nevertheless, Heur-BM lacks any approximation guarantees in terms of the blocker set cardinality, due to the non-submodularity of the optimized objective function. In light of this, we then devise a bicriteria approximation algorithm App-BM. At a high level, rather than optimizing the objective function as in Heur-BM, App-BM targets the optimization of the submodular upper bounding function, by generating a fixed number of samples. We prove that when the input error parameter μ of App-BM falls below a boundary value, the algorithm can guarantee strong approximation on the blocker set cardinality with only a minor violation of the threshold condition. However, we note that this boundary value is generally very small in practice, which causes App-BM to require an excessively large sample size and makes it mainly of theoretical interest. This limitation motivates us to further design a more efficient bicriteria approximation algorithm App-BM+, which also focuses on optimizing the upper bounding function, but improves upon App-BM by employing a more refined sampling scheme. Specifically, App-BM+ adopts a *trial-and-error* strategy to dynamically decrease μ until the pre-defined stopping conditions are satisfied. This adaptive mechanism enables App-BM+ to achieve substantially higher empirical efficiency than App-BM. As the price of enhanced efficiency, we prove that App-BM+ provides slightly weaker approximation ratios than App-BM. It is noteworthy that the underlying idea of the two approaches resembles the sandwich strategy; however, the key difference lies in the objective: our goal is to minimize the blocker set cardinality, while the sandwich strategy is devised for maximization problems. This intrinsically renders its theoretical results inapplicable to the BM setting, thus necessitating new insights and making our analysis non-trivial.

Finally, extensive experiments over 9 real-world datasets are conducted to verify the efficiency and effectiveness of our proposed techniques. The experimental results demonstrate that, for the IMIN problem, our SandIMIN algorithm achieves orders-of-magnitude speedups while maintaining competitive solution quality compared with the state-of-the-arts [60]. Moreover, for the BM problem, our algorithms are scalable and consistently deliver superior effectiveness over the existing solution [41]. Note that this paper is a significant extension of our previous work [51]. The main contributions of the paper are summarized as follows.

- For the IMIN problem, we design $(1 - 1/e - \epsilon)$-approximation algorithms to maximize the proposed submodular bounding functions. Building on these foundations, we propose SandIMIN, the first efficient approximation algorithm that provides strong data-dependent approximation guarantees.
- For the BM problem, we first propose Heur-BM, a heuristic that guarantees the threshold condition satisfaction. We then propose two bicriteria approximation algorithms, App-BM and App-BM+: App-BM achieves a tighter approximation ratio but sacrifices efficiency, while App-BM+ attains improved efficiency at the expense of slightly weaker ratios.
- We conduct extensive experiments on 9 real-world datasets to evaluate the performance of our algorithms on the IMIN and BM problems. The results demonstrate substantial improvements over the corresponding competitors across both tasks.

Roadmap The remainder of this paper is organized as follows. Section 2 introduces the formal definitions of the studied problems and revisits existing solutions. Sections 3 and 4 present the details of our solutions for the IMIN and BM problems, respectively. We conduct extensive experiments in Sect. 5. Section 6 reviews the related work, and Sect. 7 concludes the paper.

Due to the limited space, some proofs and experimental results are deferred to the online appendix [1].

2 Preliminaries

In this paper, we consider a directed graph $G = (V, E)$ with a node set V and a directed edge set E, where $|V| = n$ and $|E| = m$. Given an edge $\langle u, v\rangle \in E$, we refer to u as an incoming neighbor of v and v as an outgoing neighbor of u. Each edge $\langle u, v\rangle$ is associated with a propagation probability $p(u, v) \in [0, 1]$, representing the probability that u activates v. Table 1 summarizes the notations frequently used throughout the paper. Sect. 2.1 first introduces the diffusion model. Section 2.2 presents the formal definitions of the studied problems, Sect. 2.3 analyzes their relationship, and Sect. 2.4 reviews their existing solutions. Finally, Sect. 2.5 presents some useful sampling techniques.

Table 1 Summary of notations

Notation	Description
$G = (V, E)$	the directed graph G
k	the budget k of the IMIN problem
S, B	the seed set and the blocker set
$\mathbb{E}[I_G(S)]$	the influence spread of seed set S
$G[V']$	a subgraph of G induced by node set V'
ϕ	a random realization
$D(B)$	the decreased spread after blocking B
$D^L(\cdot)$, $D^U(\cdot)$	the lower and upper bound of $D(\cdot)$
B_L^o, B_U^o, B_1^o	the size-k optimal solutions to maximize $D^L(\cdot)$, $D^U(\cdot)$ and $D(\cdot)$
$R_\phi^-(s)$	nodes reached by s in ϕ excluding $S \cup \{s\}$
V_s'	nodes reached by s in G excluding $S \cup \{s\}$
$C^s, \mathbb{C}^s$	a CP sequence and the set of CP sequences
$L(v), \mathbb{L}$	a LRR set of v and the set of LRR sets
η	the threshold of the BM problem
B_2^o	the solution with the smallest cardinality that satisfies $D(B_2^o) \geq \eta$
B_2^*	the solution with the smallest cardinality that satisfies $D^U(B_2^*) \geq \eta$

2.1 Diffusion model

We focus on the widely-studied *independent cascade (IC)* model [5, 24, 43, 54, 55, 58]. Given a seed set $S \subseteq V$, the diffusion process of S under the IC model unfolds in discrete timestamps. At timestamp 0, only the nodes in S are activated. Each activated node will remain active in the subsequent timestamps. If a node u is activated at timestamp t, then for each of its inactive outgoing neighbor v, u has a single chance to activate v with probability $p(u, v)$ at timestamp $t + 1$. The propagation process stops when no more nodes can be activated.

2.2 Problem definition

Given a seed set S, let $I_G(S)$ be the number of active nodes in G when the stochastic propagation process stops. We define $\mathbb{E}[I_G(S)]$ as the *influence spread* of S, where the expectation is taken over the randomness of propagation. Alternatively, the diffusion process can be characterized by *live edge* procedure [24]. For any given $G = (V, E)$, by removing each $\langle u, v\rangle \in E$ with $1 - p(u, v)$ probability, the remaining subgraph is referred to as a random *realization* (denoted by ϕ) of G. Let $I_\phi(S)$ be the number of nodes reachable from S in ϕ. The influence spread $\mathbb{E}[I_G(S)]$ of S can be calculated as $\mathbb{E}[I_G(S)] = \sum_{\phi \in \Omega} I_\phi(S) \cdot p(\phi)$, where Ω is the set of all possible realizations of G, and $p(\phi)$ is the probability for realization ϕ to occur.

In this work, we consider the *node blocking* strategy to minimize the influence spread of a given seed set. When a node u is blocked, we set the probability of all edges pointing to u as 0 and refer to u as a blocker; nodes whose status transitions from active to inactive due to this operation are said to be *protected* by u. We assume that seed nodes cannot be blocked. Given a seed set $S \subseteq V$ and a blocker set $B \subseteq V \setminus S$, we denote $D_S(B) = \mathbb{E}[I_G(S)] - \mathbb{E}[I_{G[V\setminus B]}(S)]$ as the expected number of nodes protected by B, i.e., the reduced influence spread of S after blocking B, where $G[V \setminus B]$ is the subgraph of G induced by $V \setminus B$.

Problem statement 1 Given a directed graph G, a seed set S and a budget k, the *influence minimization* (IMIN) problem is to find a blocker set B_1^o with at most k nodes such that the influence spread of S is minimized after blocking B_1^o. In other words, IMIN is to identify a blocker set such that the expected number of nodes protected by B_1^o is maximized. Formally, we have

$$B_1^o = \arg\max_{B \subseteq V \setminus S, |B| \leq k} D_S(B).$$

When the context is clear, the subscript in $D_S(\cdot)$ is omitted for brevity, which we denote by $D(\cdot)$.

Xie et al. [60] have established the computational hardness of IMIN and the properties of $D(\cdot)$ under the IC model, as detailed below.

Theorem 1 ([60]) *Under the IC model, IMIN problem is NP-hard and APX-hard unless P=NP.*

Theorem 2 ([60]) *Under the IC model, for any seed set $S \subseteq V$, $D(\cdot)$ is monotonic but not submodular.*

In addition to the IMIN problem, to better support real-world applications, this work also investigates the blocker minimization problem, as defined below.

Problem statement 2 Given a directed graph G, a seed set S and a non-negative threshold η, the *blocker minimization* (BM) problem is to find a blocker set B_2^o with minimum cardinality such that the expected number of nodes protected by B_2^o is not less than η, i.e.,

$$B_2^o = \arg\min_{B \subseteq V \setminus S} |B| \text{ s.t. } D(B) \geq \eta.$$

Note that for any given S, $D(\cdot)$ is upper bounded by $\mathbb{E}[I_G(S)] - |S|$. Hence, we restrict η to the range $(0, \mathbb{E}[I_G(S)] - |S|)$. In addition, if $D(B) \geq \eta$, we can say that B meets the *threshold condition*.

In [41], Pham et al. prove the hardness of BM under the IC model, as presented below.

Theorem 3 ([41]) *Under the IC model, the BM problem is NP-hard unless P=NP.*

2.3 Relationship between IMIN and BM

The IMIN and BM problems are defined over the same set function $D(\cdot)$ but swap the roles of objective and constraint: IMIN maximizes $D(\cdot)$ under a cardinality budget, whereas BM minimizes the cardinality subject to a threshold on $D(\cdot)$. In the following, we first show that, under *exact* optimization, the two problems are mutually reducible, i.e., an exact oracle for either problem solves the other (Lemma 1). This reduction, however, is of purely theoretical interest, since both problems are NP-hard and thus intractable to solve exactly. We then show that the reducibility *breaks down under approximation*: an approximation guarantee for one problem does not transfer to the other (Lemma 2). Consequently, each problem calls for dedicated approximation algorithms.

Lemma 1 *An exact oracle for either of IMIN and BM solves the other in polynomial time.*

Proof Let $\mathsf{OPT}(k) = \max_{B \subseteq V \setminus S,\, |B| \le k} D(B)$, and let $B_2^o(t)$ be a minimum-cardinality set with $D(B_2^o(t)) \ge t$, where $|B_2^o(t)| = \infty$ if no such set exists. Both quantities are non-decreasing and satisfy

$$|B_2^o(t)| \le k \iff \mathsf{OPT}(k) \ge t, \tag{1}$$

since each side states that some blocker set of size at most k attains value at least t.

IMIN $\Rightarrow$ *BM.* Let $k^* = \min\{k : \mathsf{OPT}(k) \ge \eta\}$. By Eq. (1), $|B_2^o(\eta)| = k^*$. As $\mathsf{OPT}(\cdot)$ is monotone, binary search over k locates k^* with $O(\log n)$ IMIN queries. The set B_{k^*} returned at budget k^* satisfies $D(B_{k^*}) = \mathsf{OPT}(k^*) \ge \eta$; since no set of size below k^* attains η, it is an optimal BM solution.

BM $\Rightarrow$ *IMIN.* By Eq. (1), $\mathsf{OPT}(k) = \max\{t : |B_2^o(t)| \le k\}$, which we locate by binary search over t. By convention, all edge probabilities are rationals, and we let q be the maximum of their denominators. Then the probability of every realization, being a product of at most m such terms, is a multiple of $1/Q$ with $Q \le q^m$, and so is every attainable value of $D(\cdot)$, including $\mathsf{OPT}(k)$. It thus suffices to perform the binary search over the multiples of $1/Q$ in $(0, n]$; since the predicate $|B_2^o(t)| \le k$ is monotone in t, the search returns the largest feasible value $t^* = \mathsf{OPT}(k)$ within $O(\log(nQ)) = O(m \log q + \log n)$ BM queries. The set B returned by the BM oracle at t^* satisfies $D(B) \ge t^*$ and $|B| \le k$; hence $D(B) = t^* = \mathsf{OPT}(k)$, i.e., B is an optimal IMIN solution. □

Lemma 2 *Approximation does not transfer between IMIN and BM.* (i) *An α-approximate IMIN algorithm (which, for budget k, returns a set B with $|B| \le k$ and $D(B) \ge \alpha\, D(B_1^o)$, $0 < \alpha < 1$) yields no bounded cardinality approximation for BM.* (ii) *A β-approximate BM algorithm (which, for any threshold η, returns a B with $D(B) \ge \eta$ and $|B| \le \beta\, |B_2^o|$, $\beta > 1$) yields no non-trivial approximation guarantee for IMIN.*

Proof *IMIN* $\not\Rightarrow$ *BM.* We exhibit an α-approximate IMIN algorithm whose output stays below η until the budget far exceeds $|B_2^o|$. Consider the following instance. Let $g = \lceil 1/(1-\alpha) \rceil$, so that $(g-1)/g \ge \alpha$, fix an integer $p \ge 2$, and let $M = \lceil gp/\alpha \rceil$. The seed s has probability-1 edges to g nodes $u_1, \ldots, u_g$ and to M nodes $r_1, \ldots, r_M$, and each u_i has probability-1 edges to $p-1$ nodes reachable only through it. Blocking u_i protects p nodes while blocking any other node protects exactly 1, so $\mathsf{OPT}(k) = \min(k, g)\, p + \max(0, k-g)$. Set $\eta = gp$, which is valid as the total protectable number is $gp + M$; protecting gp nodes most cheaply requires blocking all g of the u_i, hence $|B_2^o| = g$.

Let $k^* = \min\{k : \alpha\, \mathsf{OPT}(k) > \eta - 1\}$. For each budget $k < k^*$, consider the output $B_k = \{u_1, \ldots, u_m\} \cup \{r_1, \ldots, r_t\}$, where $m = \min(k,\ g-1)$ and $t = \min(k-m,\ p-1)$, so that $|B_k| = m + t \le k$ and $D(B_k) = mp + t$. We first show that B_k is a valid α-approximate output:

- *Case 1 ($k \le g-1$).* $D(B_k) = kp = \mathsf{OPT}(k)$.
- *Case 2 ($k \ge g$, $t = p-1$).* $D(B_k) = \eta - 1 \ge \alpha\, \mathsf{OPT}(k)$, since $\alpha\, \mathsf{OPT}(k) \le \eta - 1$ for every $k < k^*$.
- *Case 3 ($k \ge g$, $t = k-g+1 < p-1$).* $D(B_k) = (g-1)p + (k-g+1) \ge \alpha gp + \alpha(k-g) = \alpha\, \mathsf{OPT}(k)$, due to $(g-1)/g \ge \alpha$ and $\alpha < 1$.

Yet, $D(B_k) \le (g-1)p + (p-1) = \eta - 1 < \eta$ in every case, i.e., B_k never meets the threshold. This indicates that, within this budget range, an α-approximate IMIN algorithm cannot produce BM-feasible solution.

For each budget $k \ge k^*$, consider the optimal size-k IMIN solution B_1^o, which meets the threshold as $D(B_1^o) = \mathsf{OPT}(k) \ge \mathsf{OPT}(g) = \eta$, noting that $k^* > g$ since $\alpha\, \mathsf{OPT}(g) = \alpha gp \le gp - 1$ by $g(1-\alpha) \ge 1$ and $p \ge 2$. It remains to quantify k^*. Since $\mathsf{OPT}(k) = gp + (k-g)$ for $k \ge g$, the defining condition $\alpha\, \mathsf{OPT}(k) > \eta - 1$ becomes $\alpha(gp + k - g) > gp - 1$, which gives $k^* = g + \lceil (gp(1-\alpha) - 1)/\alpha \rceil$. Thus k^* grows without bound as p increases, whereas $|B_2^o| = g$ stays constant. Hence, an α-approximate IMIN algorithm yields no bounded cardinality approximation for BM.

BM $\not\Rightarrow$ *IMIN.* We exhibit a β-approximate BM algorithm whose outputs, when used for IMIN under budget k, attain only a vanishing fraction of $\mathsf{OPT}(k)$. Consider the following instance. Let $k = \lceil 1/(\beta - 1) \rceil$, the smallest integer with $\beta k \ge k + 1$, and fix an integer $M \ge \beta k$. The seed s has probability-1 edges to k nodes $v_1, \ldots, v_k$, and each v_i has probability-1 edges to every node of $\{w_1, \ldots, w_M\}$. Since each w_j remains reachable through any unblocked v_i, a blocker set B not containing all of $v_1, \ldots, v_k$ protects only

itself, whereas blocking all k of the v_i protects everything. Hence $\mathsf{OPT}(k-1) = k-1$ and $\mathsf{OPT}(k) = k+M$, and the BM optimum at any threshold t is $|B_2^o(t)| = \min(\lceil t \rceil, k)$. For each threshold t, consider the output B_t defined below:

- *Case 1* ($\lceil t \rceil \leq k-1$). $B_t = \{w_1, \ldots, w_{\lceil t \rceil}\}$, which is an optimal BM solution.
- *Case 2* ($\lceil t \rceil \geq k$). $B_t = \{v_1, \ldots, v_k\} \cup \{w_1, \ldots, w_{\lfloor \beta k \rfloor - k}\}$, so that $D(B_t) \geq t$ and $|B_t| = \lfloor \beta k \rfloor \leq \beta k = \beta\, |B_2^o(t)|$.

Both cases are valid β-approximate outputs. Now use this algorithm for budget-k IMIN, where an output is usable only if its cardinality is at most k. In Case 2, $|B_t| = \lfloor \beta k \rfloor \geq k+1 > k$, so every threshold with $\lceil t \rceil \geq k$ overflows the budget; in Case 1, the obtained value is only $D(B_t) = \lceil t \rceil \leq k-1$. The best IMIN value obtainable from this algorithm is therefore $\mathsf{OPT}(k-1) = k-1$, whereas $\mathsf{OPT}(k) = k+M$. The approximation ratio is thus at most $(k-1)/(k+M)$, which vanishes as M grows. Hence, a β-approximate BM algorithm yields an IMIN approximation ratio arbitrarily close to 0. □

2.4 Existing solutions revisited

2.4.1 Existing IMIN solutions

Given the hardness of the IMIN problem, existing IMIN solutions [52, 60] are all based on the greedy strategy. In the following, we first outline the greedy framework for IMIN, and then detail the state-of-the-art algorithms, along with their empirical and theoretical limitations.

Greedy framework Let $D(u|B) = D(B \cup \{u\}) - D(B)$ be the marginal gain of adding u to B w.r.t. $D(\cdot)$. In a nutshell, the greedy framework starts from an empty blocker set B. The subsequent part of the algorithm consists of k iterations. At each iteration, it iteratively selects the node v from $V \backslash S$ that leads to the largest $D(v|B)$ and adds it into B. After selecting one blocker, the probability of all edges pointing to it is set as 0. Due to $D(B) = \mathbb{E}[I_G(S)] - \mathbb{E}[I_{G[V\backslash B]}(S)]$, one feasible way for calculating $D(B)$ is to compute $\mathbb{E}[I_G(S)]$. However, the computation of $\mathbb{E}[I_G(S)]$ is proved as #P-hard [7], which means that the computation of $D(B)$ is also #P-hard. In [52], Wang et al. use Monte-Carlo simulations to estimate the influence spread $\mathbb{E}[I_G(S)]$ and adopt the greedy framework. It can solve IMIN effectively but due to the inefficiency of Monte-Carlo simulations, it incurs significant computation overhead.

State-of-the-arts In comparison with the Monte-Carlo based estimation method under the greedy framework, the state-of-the-art solutions [60] optimize the estimation for $D(\cdot)$. Instead of estimating the influence spread of S, Xie et al. [60] explicitly estimate the reduced influence spread based on *dominator tree* (DT) [2, 33], with the definition shown below.

Definition 1 *(Dominator)* Given a realization ϕ and a source s, a node u is called a dominator of a node v if and only if all paths from s to v pass through u.

Definition 2 *(Immediate Dominator)* Given a realization ϕ and a source s, a node u is said to be an immediate dominator of a node v, denoted as $idom(v) = u$, if and only if u dominates v and every other dominator of v dominates u.

Definition 3 *(Dominator Tree)* Given a realization ϕ and a source s, the dominator tree of ϕ is induced by the edge set $\{\langle idom(u), u\rangle : u \in V \backslash \{s\}\}$ with root s.

As presented, each DT has only a single root. Xie et al. [60] first propose to replace the seed set S with a unified node s that exhibits an equivalent non-seed influence spread under the IC model, i.e., $\mathbb{E}[I_G(S)] - |S| = \mathbb{E}[I_G(s)] - 1$. In particular, for each node $u \in V \backslash S$, if there are h distinct seed nodes pointing to node u, and each edge has a probability p_i ($1 \leq i \leq h$), they remove all edges from the seed nodes to node u, and add an edge from node s to node u with the probability $1 - \prod_{i=1}^{h}(1-p_i)$. Besides, Xie et al. [60] prove that for each $u \in V \backslash S$, the expected number of protected nodes by u (i.e., $D(u)$) equals the expected size of the subtree rooted at u in DT. With these results, the estimation procedure in [60] runs as follows.

- Generate a number of random realizations $\mathcal{G}$ from G.
- For each realization $\phi \in \mathcal{G}$, apply Lengauer-Tarjan algorithm [29] to construct the DT of ϕ, rooting at s. For each node $v \in V \backslash S$, measure the size $c_\phi(v)$ of subtree with the root v in DT.
- For each node $v \in V \backslash S$, the estimated value $\hat{D}_s(v)$ of $D_s(v)$ is the average value of $c_\phi(v)$ in all the generated realizations, i.e., $\hat{D}_s(v) = \sum_{\phi \in \mathcal{G}} c_\phi(v)/|\mathcal{G}|$.

Example 1 An example of the estimation procedure is illustrated, where we only generate one realization for simplicity. Figure 1(a) shows a directed graph $G = (V, E)$, where v_0 is the seed node. The number associated with each edge denotes its propagation probability. Figure 1(b) shows a random realization ϕ of G. The DT rooted at v_0 of ϕ is presented in Fig. 1(c). Then, we have $\hat{D}_s(v_1) = \hat{D}_s(v_2) = \hat{D}_s(v_5) = \hat{D}_s(v_6) = 1$, $\hat{D}_s(v_3) = 3$ by measuring the size of corresponding subtree in DT. As v_4 cannot be activated in ϕ, $\hat{D}_s(v_4) = 0$.

By combining this advanced estimation procedure with the greedy framework, Xie et al. [60] first propose the Advanced-Greedy (AG) approach. Nevertheless, they observe that some critical nodes may be missed by AG, e.g., some outgoing neighbors of the seed nodes. Consider Fig. 1(a), suppose $k = 2$. v_0 has two outgoing neighbors and if we directly select these two nodes as blockers, v_0 cannot activate any node. However, if we block the blockers returned

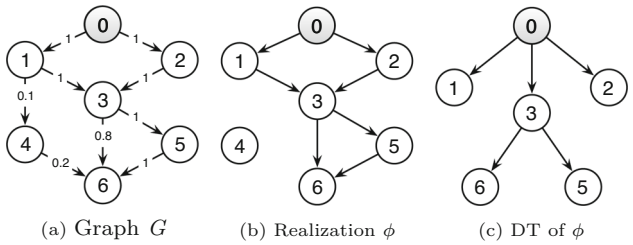


Fig. 1 A toy directed graph

by AG (i.e., v_3 and v_1), v_2 cannot be protected. To address this issue, Xie et al. [60] further propose Greedy-Replace (GR), which generally consists of two stages. In the first stage, the outgoing neighbors of seed nodes are stored in the candidate set. They iteratively select the node v from the candidate set with the largest $\hat{D}_s(v|B)$ and add it into B, until $|B| = \min\{d_s^{out}, k\}$, where d_s^{out} denotes the number of nodes in the candidate set. In the second stage, they consider processing the blockers in B according to the reverse order of their insertion order. For each blocker in B, they first remove it from B, called the replaced node. Then they select v from $V \setminus S$ with the largest $\hat{D}_s(v|B)$ and add it into B, called the current best blocker. If the replaced node is the current best blocker, they return B directly. Otherwise, continue the replacement process.

Compared with AG, GR often achieves better result quality. However, AG is typically more efficient than GR, as the latter involves a two-stage selection process and may require multiple rounds of replacement in the second stage before termination.

Limitations Despite the efficiency of dominator tree based estimation method, AG and GR still incur significant computation overhead in practice. This is because once AG/GR selects a node as a blocker, that node would be removed from the graph. As a result, the realizations from the preceding round cannot be reused, and both algorithms must regenerate them (in the updated graph) along with the corresponding DTs, incurring substantial time overhead when k is large.

On the other hand, due to the non-submodularity of $D(\cdot)$, all the greedy-based algorithms (including AG and GR) cannot guarantee a $(1 - 1/e)$ approximation ratio [37]; additionally, no theoretical analysis has been conducted for them. Based on the notions of curvature and submodularity ratio [4], here we analyze the approximation guarantee for the greedy strategy on the non-submodular IMIN objective. The submodularity ratio serves as a metric to assess how closely the objective approximates being submodular. Formally, for all $A, B \subset V$, the submodularity ratio of $D(\cdot)$ is the largest scalar ψ such that,

$$\sum_{\omega \in A \setminus B}(D(B \cup \{\omega\}) - D(B)) \geq \psi \cdot (D(B \cup A) - D(B)). \quad (2)$$

Consider a directed graph $G = (V, E)$ with $V = \{v_0, v_1, \ldots, v_{n-1}\}$ and $E = \{\langle v_0, v_1\rangle, \langle v_0, v_2\rangle, \langle v_1, v_3\rangle, \langle v_2, v_3\rangle, \langle v_3, v_4\rangle, \langle v_3, v_5\rangle, \ldots, \langle v_3, v_{n-1}\rangle\}$. The probability on each edge is set to 1 and v_0 is the seed node. When $B = \emptyset$ and $A = \{v_1, v_2\}$, the left side of Eq. (2) is equal to 2, and the right side of Eq. (2) is equal to $n - 1$, i.e., $2 \geq \psi \cdot (n - 1)$. As n gradually becomes larger, the submodularity ratio ψ of the IMIN objective approaches 0 infinitely, which in turn causes the approximation ratio of the greedy method to degrade to 0 [4]. Thus, the state-of-the-art IMIN algorithms cannot provide any non-trivial guarantees. In this paper, we conduct the first attempt to fill these gaps by designing SandIMIN, an efficient approximation algorithm for IMIN with strong performance guarantees.

2.4.2 Existing BM solutions

To the best of our knowledge, the only prior study of the BM problem is by Pham et al. [41], in which a scalable greedy-based heuristic, referred to as Base, is introduced. In what follows, we first abstract the greedy framework (which we call Thre-Greedy) utilized by Pham et al., and then detail their algorithm for BM.

Algorithm 1: Thre-Greedy (G, η, f)

```
1 B ← ∅;
2 while f(B) < η do
3 |   u ← arg max_{v∈V\B} f(v|B);
4 |   B ← B ∪ {u};
5 return B;
```

As outlined in Algorithm 1, given a threshold η and a function f, Thre-Greedy iteratively selects the element u with the largest $f(u|B)$ and adds it to B, until $f(B) \geq \eta$ (Lines 2-4). Here $f(u|B) = f(B \cup \{u\}) - f(B)$ represents the marginal gain of adding u to B w.r.t. $f(\cdot)$. Thre-Greedy is practically effective and has been widely applied to tackle various optimization problems (e.g., set cover [57]). For any submodular and monotonic function f, Thre-Greedy provides the following non-trivial approximation guarantees [19, 63].

Theorem 4 ([19, 63])

Let B be the solution returned by Thre-Greedy(G, η, f), $B_i \subseteq B$ be the set of the first i selected elements, and $\tau = \max_{v \in V} f(v|B_{|B|-1})$. Let B^o be the minimum-cardinality solution satisfying $f(B^o) \geq \eta$. If f is submodular, monotonic and upper-bounded by n, we then have $f(B) \geq \eta$ and $|B| \leq \min\{\beta_1, \beta_2\} \cdot |B^o|$, where $\beta_1 = 1 + \ln \frac{\eta}{\tau |B^o|}$ and $\beta_2 = \lceil \ln \frac{n\eta}{n-\eta} \rceil + \frac{1}{|B^o|}$.

To address the BM problem, Pham et al. [41] propose Base, an adaptation of Thre-Greedy that optimizes the estimation function $\tilde{D}(\cdot)$ of $D(\cdot)$. That is, beginning with an

empty blocker set B, Base iteratively selects the node u that yields the maximum marginal gain $\tilde{D}(u|B)$ and adds it to B, until $\tilde{D}(B) \geq \eta$. The estimator $\tilde{D}(\cdot)$ is computed via the tree structure. Specifically, Base first generates a certain number of random realizations. For each realization ϕ, it constructs a tree T_ϕ using the DFS algorithm starting from s, ensuring that each visited node is assigned a unique incoming neighbor corresponding to the node that first reaches it. Here s is the unified seed of S, constructed consistently with the method in [60], as we introduced in Sect. 2.4. For any blocker set B, the authors [41] define $\tilde{D}(B) = \frac{1}{|\mathcal{L}|}\sum_{T_\phi \in \mathcal{L}} I_{T_\phi}(B)$, where $\mathcal{L}$ denotes the set of constructed trees and $I_{T_\phi}(B)$ denotes the number of nodes reachable from B on T_ϕ.

To avoid regenerating all trees after each new insertion into B, Base stores the structural information of each $T_\phi \in \mathcal{L}$, i.e., the child and ancestor relations of its contained nodes. This operation, however, incurs substantial memory overhead, particularly when the seed s is influential. To enable the practical implementation, the authors [41] empirically adopt a relatively small sample size (i.e., 1000) for Base.

Limitations Base is highly efficient, as it generates only a small number of trees and employs a greedy selection procedure that is typically fast. However, it can be verified that the estimator $\tilde{D}(\cdot)$ is biased with respect to $D(\cdot)$. As a consequence, Base provides no theoretical guarantees and may produce suboptimal outcomes. In this paper (Sect. 4), we first design Heur-BM, which satisfies the threshold condition by utilizing an unbiased estimator of $D(\cdot)$. Building upon the theoretical results presented in Theorem 4, we further devise two bicriteria approximation algorithms (App-BM and App-BM+) that offer provable performance guarantees on the cardinality of the returned blocker set.

2.5 Sampling techniques

This section presents two sampling techniques that will be utilized in the subsequent algorithm design.

RIS approach Reverse Influence Sampling (RIS) [5] is widely used for influence spread estimation. This technique is based on the concept of random reverse reachable (RR) set, which is denoted by R and generated by the following operations.

- Sample a source node v uniformly at random from V and construct a random realization ϕ of G.
- Identify the set of nodes that can reach v under ϕ and store them to R.

Given a random RR set R whose source node is v, for each node $u \in V$, the probability that u appears in R equals the probability that u can activate v. Intuitively, if a seed set S is highly influential, then there is a high chance that some nodes in S appear in a random RR set. In [5], Borgs et al. establish an unbiased estimator for the influence spread $\mathbb{E}[I_G(S)]$ of S.

Lemma 3 *Let $\mathcal{R}$ be a collection of random RR sets and $Cov_{\mathcal{R}}(S) = \sum_{R\in\mathcal{R}} \min\{|S \cap R|, 1\}$ be the coverage of S in $\mathcal{R}$, we have*

$$\mathbb{E}[I_G(S)] = n \cdot \mathbb{E}[Cov_{\mathcal{R}}(S)/|\mathcal{R}|], \tag{3}$$

where the expectation is over the randomness of $\mathcal{R}$.

With this result, by generating a sufficient number of random RR sets, we can accurately estimate $\mathbb{E}[I_G(\cdot)]$.

Stopping rule for sampling The stopping rule algorithm [65] is to estimate a single random variable X in the range of $[a, b]$ with $\mathbb{E}[X] = \mu_X$. Specifically, it repeatedly generates samples until the sum of the observed values exceeds $\mathcal{T}(a, b, \gamma, \delta)$, as defined below.

$$\mathcal{T}(a, b, \gamma, \delta) = \frac{2}{\gamma^2}(b-a)\,(1+\gamma)\left(\frac{b-a}{b} + \frac{\gamma}{3}\right)\ln\frac{2}{\delta}. \tag{4}$$

The estimator $\hat{\mu}_X = \mathcal{T}(a, b, \gamma, \delta)/\theta$ (where θ is the number of samples generated) is an (γ, δ)-estimate of μ_X, i.e., $\Pr\big[(1-\gamma)\mu_X \leq \hat{\mu}_X \leq (1+\gamma)\mu_X\big] \geq 1-\delta$. This guarantee is derived by a martingale argument together with an Azuma–Hoeffding-type concentration inequality.

3 Solutions for influence minimization

3.1 Sandwich approximation strategy

To solve the IMIN problem, we propose an efficient approximation algorithm SandIMIN based on the sandwich framework [34, 56]. In Sect. 3.1.1, we first give the general framework of SandIMIN. Then we develop the lower and upper bounding functions of our objective function in Sect. 3.1.2 and 3.1.3, respectively.

Before presenting the details, we first propose a simple technique for transforming multiple seeds S into a unified seed s, such that $\mathbb{E}[I_G(s)] = \mathbb{E}[I_G(S)] + 1$. In particular, we create a virtual seed node s and introduce the edges with the propagation probability of 1 from s to every seed node $v \in S$. Under such a setting, it is clear that minimizing $\mathbb{E}[I_G(s)]$ is equivalent to minimizing $\mathbb{E}[I_G(S)]$, where the blocker set is restricted to be selected from $V\backslash S$. Note that s is a virtual node and does not belong to V. Different from the seed merging technique stated in Sect. 2.4, this approach avoids the pre-computation of $1 - \prod_{i=1}^{h}(1-p_i)$ for each $u \in V\backslash S$.

Algorithm 2: SandIMIN($G, S, k, \gamma, \epsilon, \delta$)

Input : The graph $G = (V, E)$, the seed set S, the budget k and the error parameters γ, ϵ, δ.
Output : The blocker set B.

1 $B_L \leftarrow$ App-LB($G, S, k, \gamma, \epsilon, \delta$); /* Section 3.3.1 */;
2 $B_U \leftarrow$ App-UB($G, S, k, \epsilon, \delta$); /* Section 3.3.2 */;
3 $B_R \leftarrow$ Heur-IMIN(G, S, k); /* Section 3.3.3 */;
4 $\hat{I}_{G[V\setminus\cdot]}(s) \leftarrow$ the (γ, δ)-estimate of $\mathbb{E}[I_{G[V\setminus\cdot]}(s)]$;
5 $B \leftarrow \arg\min_{B^* \in \{B_L, B_U, B_R\}} \hat{I}_{G[V\setminus B^*]}(s)$;
6 **return** B;

3.1.1 Overview of SandIMIN

Let D^L, D^U be the non-negative, monotonic and submodular set functions defined on V, i.e., $D^L : 2^V \rightarrow \mathbb{R}_{\geq 0}$ and $D^U : 2^V \rightarrow \mathbb{R}_{\geq 0}$, such that $\forall B \subseteq V \setminus S$, $D^L(B) \leq D(B) \leq D^U(B)$. Algorithm 2 presents the basic workflow of SandIMIN. To begin with, it invokes the App-LB (resp. App-UB) approach to obtain B_L (resp. B_U), which offers $(1 - 1/e - \epsilon)$-approximation for maximizing $D^L(\cdot)$ (resp. $D^U(\cdot)$) with at least $1 - \delta$ probability. It then invokes Heur-IMIN to obtain B_R, a heuristic solution for the IMIN problem. Finally, it returns the blocker set $B^* \in \{B_L, B_U, B_R\}$ with the smallest (γ, δ)-estimate of $\mathbb{E}[I_{G[V\setminus B^*]}(s)]$.

The detailed implementation of App-LB, App-UB and Heur-IMIN will be presented in Sect. 3.3. Let B_L^o (resp. B_U^o) be the size-k optimal solutions to the tasks of maximizing $D^L(\cdot)$ (resp. $D^U(\cdot)$). The following theorem shows the approximation guarantees of SandIMIN.

Theorem 5 *SandIMIN returns B that satisfies*

$$\Pr\left[D(B) \geq \tau(1 - 1/e - \epsilon)D(B_1^o)\right] \geq 1 - 3\delta,$$

where $\tau = \max\left\{\frac{D^L(B_L^o)}{D(B_1^o)}, \frac{D(B_U)}{D^U(B_U)}\right\} \cdot \frac{1-\gamma}{1+\gamma}$ *is a data-driven parameter and B_1^o is the size-k optimal IMIN solution.*

3.1.2 Development of $D^L(\cdot)$

The non-submodularity of the function $D(\cdot)$ arises from the *combination effect* among nodes in the blocker set B. That is, to prevent a node from being activated by s, we need to simultaneously block two or more nodes. For example, reconsidering Fig. 1, to protect v_3, we need to simultaneously block v_1 and v_2. Motivated by this, we disregard the combination effect to obtain a submodular lower bound. Specifically, we only consider nodes that can be protected by blocking only one node in G. The lower bound $D^L(B)$ of $D(B)$ is defined below.

$$D^L(B) = \sum_{\phi \in \Omega} p(\phi) \cdot |\cup_{v \in B} N_\phi(v)|, \quad (5)$$

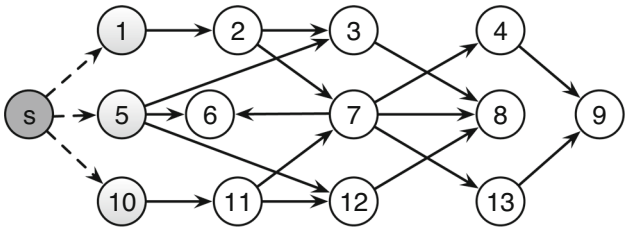


Fig. 2 Example of the bounds

where $N_\phi(v)$ is the set of nodes protected by v in ϕ. Lemma 4 then proves the submodularity of $D^L(\cdot)$

Lemma 4 *$D^L(\cdot)$ is monotone nondecreasing and submodular under the IC model.*

Besides, as $D^L(\emptyset) = \mathbb{E}[I_G(s)]$ and computing $\mathbb{E}[I_G(s)]$ is #P-hard [7], the computation of $D^L(\cdot)$ is #P-hard.

3.1.3 Development of $D^U(\cdot)$

We say a path is *blocked* if it contains at least one node from B. For each v activated by s, v can be protected if all the paths from s to v are blocked. Intuitively, we can obtain the upper bound of $D(\cdot)$ by relaxing the condition for nodes to be protected. We call that v can be *alternative-protected* if there exists one path from s to v that is blocked. Given a realization ϕ, we define $R_\phi^-(s)$ as the set of nodes (excluding $S \cup \{s\}$) reachable from s in ϕ, ϕ^s as the subgraph of ϕ induced by $R_\phi^-(s)$, and $R_{\phi^s}(B)$ as the set of nodes reachable from B in ϕ^s. It then follows that under ϕ, B can alternative-protect all nodes in $R_{\phi^s}(B)$. Following this insight, we define an upper bound $D^U(B)$ of $D(B)$ below.

$$D^U(B) = \sum_{\phi \in \Omega} p(\phi) \cdot |R_{\phi^s}(B)|. \quad (6)$$

We remark that maximizing $D^U(\cdot)$ can be regarded as an instance of the influence maximization problem [24]. Hence, $D^U(\cdot)$ satisfies the key properties of the influence spread function-monotonicity, submodularity, and #P-hardness of exact computation.

Example 2 An example is provided to illustrate our bounding functions. As shown in Fig. 2, the seed set $S = \{v_1, v_5, v_{10}\}$. We first create a unified seed s and introduce three edges, $\langle s, v_1\rangle, \langle s, v_5\rangle, \langle s, v_{10}\rangle$, with the propagation probability of 1. Let $B = \{v_3, v_7, v_{12}\}$. We then have $D^L(B) = |\{v_3, v_4, v_7, v_9, v_{12}, v_{13}\}| = 6$, $D(B) = |\{v_3, v_4, v_7, v_8, v_9, v_{12}, v_{13}\}| = 7$, and $D^U(B) = |\{v_3, v_4, v_6, v_7, v_8, v_9, v_{12}, v_{13}\}| = 8$.

3.2 Bounding functions estimation

To achieve Lines 1-2 in Algorithm 2, we first need to compute $D^L(\cdot)$ and $D^U(\cdot)$. However, their computation are both

#P-hard. The RIS approach for influence spread estimation (introduced in Sect. 2.5) cannot be applied to estimate our bounding functions. Specifically, we only need to consider the nodes activated by s, as only these nodes require protection and can contribute to the bounding functions. Besides, the number of these nodes is unknown due to the randomness of propagation. We address these challenges by developing two new sampling-based methods to estimate $D^L(\cdot)$ and $D^U(\cdot)$, as detailed in Sect. 3.2.1 and Sect. 3.2.2.

3.2.1 Lower bounding function estimation

We first introduce the notion of CP sequence, a core component in our estimation method for $D^L(\cdot)$.

Definition 4 *(CP Set & CP Sequence)* Given a directed graph $G = (V, E)$, a random realization ϕ generated on G, a seed set S with its unified seed s, for any $v \in V \backslash S$, a random common path (CP) set of v, denoted by $C_\phi(s, v)$, is the set of common nodes (exclude S and s) on all paths from s to v in ϕ, i.e.,

$$C_\phi(s, v) = \left\{ u \in \bigcap_{i=1}^{j} P_i(s, v) \;\middle|\; u \notin S \cup \{s\} \right\}$$

where $P_i(s, v)$ denotes the set of nodes on the i-th path from s to v in ϕ, and j is the total number of such paths. A random common path (CP) sequence, denoted by C^s_ϕ, is the set of random CP sets of the nodes in $R^-_\phi(s)$, i.e., $C^s_\phi = \{C_\phi(s, v) \mid v \in R^-_\phi(s)\}$. Note that all CP sets in C^s_ϕ are constructed under the same random ϕ.

When the context is clear, the subscript ϕ in $C_\phi(s, v)$ and C^s_ϕ can be dropped for simplicity.

Example 3 Reconsider Fig. 1, where (a) presents a graph G with seed node v_0, and (b) demonstrates a random realization ϕ of G. We first illustrate the random CP set of v_6 as follows. We can find that there are four paths between v_0 and v_6, i.e., $P_1(v_0, v_6) = \{v_0, v_1, v_3, v_6\}$, $P_2(v_0, v_6) = \{v_0, v_2, v_3, v_6\}$, $P_3(v_0, v_6) = \{v_0, v_1, v_3, v_5, v_6\}$ and $P_4(v_0, v_6) = \{v_0, v_2, v_3, v_5, v_6\}$. The random CP set of v_6 is the common nodes on four paths but excluding v_0, i.e., $C(v_0, v_6) = \{v_3, v_6\}$. A random CP sequence is the set of random CP sets of nodes in $R^-_\phi(v_0) = \{v_1, v_2, v_3, v_5, v_6\}$.

Given a set $\mathbb{C}^s$ of random CP sequences and a blocker set B, define $Cov_{\mathbb{C}^s}(B)$ be the coverage of B in $\mathbb{C}^s$, i.e.,

$$Cov_{\mathbb{C}^s}(B) = \sum_{C^s \in \mathbb{C}^s} \sum_{C(s,v) \in C^s} \min\{|B \cap C(s, v)|, 1\}.$$

Lemma 5 establishes an unbiased estimator of $D^L(B)$.

Algorithm 3: CP-Gen(G, S)

```
1  C^s_φ ← ∅;
2  φ ← a random realization φ generated on G;
3  Apply the Lengauer-Tarjan algorithm on φ to build a DT that is
   rooted at s;
4  for each u ∈ R^-_φ(s) with the order of DFS from s do
5      if idom[u] ∉ S then
6          C_φ(s, u) ← C_φ(s, idom[u]) ∪ {u};
7      else
8          C_φ(s, u) ← {u};
9      C^s_φ ← C^s_φ ∪ {C_φ(s, u)};
10 return C^s_φ;
```

Lemma 5 *Given a seed set $S \subseteq V$, and the set $\mathbb{C}^s$ of random CP sequences, for any $B \subseteq V \backslash S$, we have*

$$D^L(B) = \mathbb{E}\left[Cov_{\mathbb{C}^s}(B)/|\mathbb{C}^s|\right], \tag{7}$$

where the expectation is over the randomness of $\mathbb{C}^s$.

This indicates that an accurate estimate of $D^L(B)$ can be obtained by generating sufficient random CP sequences, which in turn necessitates an advanced CP sequence generation approach. One feasible idea is to generate random CP sets separately for each $v \in R^-_\phi(s)$, where each CP set is constructed by exhaustively enumerating all paths from s to v. This approach, however, is rather computationally expensive, as the number of paths between two nodes can be prohibitively large, and each CP sequence may include many CP sets.

To alleviate this issue, we devise a simple yet more efficient method, CP-Gen, based on the notion of DT (Definition 3). As presented in Algorithm 3, we first generate a random realization ϕ on G and construct the corresponding DT using the Lengauer-Tarjan algorithm [29], where the immediate dominator $idom[u]$ is recorded for each $u \in R^-_\phi(s)$ (Lines 2-3). By the definition of DT, for each $v \in V \backslash S$, all the nodes on the path from s to v in DT, excluding $S \cup \{s\}$, constitute $C_\phi(s, v)$. Then, leveraging the tree structure of the DT (i.e., each node has exactly one incoming edge), we perform a single DFS traversal from s in ϕ, during which the CP set of each $v \in R^-_\phi(s)$ is inherited from the CP set of $idom[v]$, i.e., v's incoming neighbor in the DT. All generated CP sets are incrementally stored in the initially empty set C^s_ϕ (Lines 4-9).

3.2.2 Upper bounding function estimation

A sampling-based method is also designed to estimate $D^U(\cdot)$, building upon the concept of the LRR set.

Definition 5 *(LRR Set)* Given a directed graph $G = (V, E)$, a random realization ϕ generated on G, a seed set S with its

Algorithm 4: LRR-Gen(G, S)

1 $\phi \leftarrow$ a random realization ϕ generated on G;
2 $v \leftarrow$ a node selected uniformly at random from V'_s;
3 **if** $v \in R^-_\phi(s)$ **then**
4 Conduct a BFS from v along the reverse edges in ϕ_s, and store all visited nodes in the set L;
5 **else** $L \leftarrow \emptyset$;
6 **return** L;

unified seed s, and a node v selected uniformly at random from V'_s (the set of nodes reachable from s in G, excluding $S \cup \{s\}$), a random local reverse reachable (LRR) set, denoted by L, is defined as the set of nodes that can reach v in ϕ^s, where ϕ^s is the subgraph of ϕ induced by $R^-_\phi(s)$.

Define $\mathbb{L}$ be a set of random LRR sets, and $Cov_{\mathbb{L}}(B) = \sum_{L\in\mathbb{L}} \min\{|B \cap L|, 1\}$ be the coverage of B in $\mathbb{L}$. Lemma 6 provides an unbiased estimator for $D^U(B)$.

Lemma 6 *Given a seed set $S \subseteq V$, and the set $\mathbb{L}$ of random LRR sets, for any $B \subseteq V \backslash S$, we have*

$$D^U(B) = |V'_s| \cdot \mathbb{E}[Cov_{\mathbb{L}}(B)/|\mathbb{L}|], \tag{8}$$

where the expectation is over the randomness of $\mathbb{L}$.

The random LRR set generation method, LRR-Gen, is shown in Algorithm 4. We first generate a random realization ϕ on G and randomly select a node v from V'_s (Lines 1-2). If v is reachable from s in ϕ, a BFS is performed from v along the reverse edges in ϕ^s to identify L. Otherwise, L is set to be an empty set (Lines 3-5).

3.3 Detailed implementation of SandIMIN

Building upon the proposed bounding function estimation approaches, this section elaborates on each component of SandIMIN. We first present efficient algorithms with $(1 - 1/e - \epsilon)$-approximation to maximize the lower (resp. upper) bounding functions, in Sect. 3.3.1 (resp. Section 3.3.2). We then propose a heuristic algorithm to solve IMIN in Sect. 3.3.3.

3.3.1 Lower bounding function maximization

Even though the unbiased estimator has been established for $D^L(\cdot)$, it is not sufficient to obtain a good approximate solution for its maximization. A key challenge lies in computing an appropriate sample size that balances computational efficiency and estimation accuracy: too few samples may lead to inaccurate results, while too many can impose significant computational overhead. In view of this, one may resort to OPIM-C [43], the state-of-the-art approach for influence maximization. In a nutshell, OPIM-C first generates two independent collections of samples $\mathcal{R}_1$ and $\mathcal{R}_2$ and then identifies a candidate solution by using the greedy strategy on $\mathcal{R}_1$. Afterward, it assesses whether the solution satisfies the stopping criterion defined by $\mathcal{R}_2$. If the criterion is met, the solution is returned; otherwise, the sizes of $\mathcal{R}_1$ and $\mathcal{R}_2$ are doubled with new samples, and the above steps are repeated until a qualified solution is obtained. In particular, the stopping criterion of OPIM-C is $\mathcal{M}(S) \geq 1 - 1/e - \epsilon$, where S is the candidate solution and $\mathcal{M}(\cdot)$ is the function determined by two martingale-based concentration bounds [46].

However, we remark that OPIM-C cannot be trivially adapted to our problem. The main reason is that OPIM-C relies

Algorithm 5: App-LB(G, S, k, γ, ϵ, δ)

Input : The graph $G = (V, E)$, the seed set S, the budget k and parameters γ, ϵ, δ.
Output : The blocker set B_L.

1 $ON \leftarrow$ the set of outgoing neighbors of S;
2 **if** $|ON| \leq k$ **then**
3 **return** $B_L = ON$;
4 $\hat{I}_G(s) \leftarrow$ the $(\gamma, \frac{\delta}{6})$-estimate of $\mathbb{E}[I_G(s)]$;
5 $\text{OPT}^L \leftarrow$ the lower bound of $D^L(B^o_L)$;
6 $\theta_{\max} \leftarrow \frac{2\hat{I}_G(s)\left((1-\frac{1}{e})\sqrt{\ln\frac{12}{\delta}}+\sqrt{(1-\frac{1}{e})(\ln\binom{n-|S|}{k}+\ln\frac{12}{\delta})}\right)^2}{(1-\gamma)\epsilon^2\text{OPT}^L}$;
7 $\theta_0 \leftarrow \theta_{\max} \cdot (1-\gamma)\epsilon^2\text{OPT}^L/\hat{I}_G(s)$;
8 $i_{\max} \leftarrow \lceil \log_2 \frac{\theta_{\max}}{\theta_0} \rceil$;
9 Generate two sets $\mathbb{C}^s_1$, $\mathbb{C}^s_2$ of θ_0 random CP sequences;
10 $a_1 \leftarrow \ln \frac{3i_{max}}{\delta}$, $a_2 \leftarrow \ln \frac{3i_{max}}{\delta}$;
11 **for** $i \leftarrow 1$ *to* $i_{\max}$ **do**
12 $B_L \leftarrow$ Max-Coverage($\mathbb{C}^s_1, k$);
13 $\sigma^L(B_L) \leftarrow 0$;
14 **if** $Cov_{\mathbb{C}^s_2}(B_L) \cdot (1-\gamma)/\hat{I}_G(s) \geq 5a_1/18$ **then**
15 $\sigma^L(B_L) \leftarrow \left((\sqrt{\frac{Cov_{\mathbb{C}^s_2}(B_L)(1-\gamma)}{\hat{I}_G(s)} + \frac{2a_1}{9}} - \sqrt{\frac{a_1}{2}})^2 - \frac{a_1}{18}\right)\frac{1}{|\mathbb{C}^s_2|}$;
16 **else if** $Cov_{\mathbb{C}^s_2}(B_L) \cdot (1+\gamma)/\hat{I}_G(s) \leq 5a_1/18$ **then**
17 $\sigma^L(B_L) \leftarrow \left((\sqrt{\frac{Cov_{\mathbb{C}^s_2}(B_L)(1+\gamma)}{\hat{I}_G(s)} + \frac{2a_1}{9}} - \sqrt{\frac{a_1}{2}})^2 - \frac{a_1}{18}\right)\frac{1}{|\mathbb{C}^s_2|}$;
18 $Cov^u_{\mathbb{C}^s_1}(B^o_L) \leftarrow \min_{0\leq i\leq k}\left(Cov_{\mathbb{C}^s_1}(B_i) + \sum_{v\in maxMC(B_i,k)} Cov_{\mathbb{C}^s_1}(v \mid B_i)\right)$;
19 $\sigma^U(B^o_L) \leftarrow \left(\sqrt{\frac{Cov^u_{\mathbb{C}^s_1}(B^o_L)(1+\gamma)}{\hat{I}_G(s)} + \frac{a_2}{2}} + \sqrt{\frac{a_2}{2}}\right)^2 \cdot \frac{1}{|\mathbb{C}^s_1|}$;
20 **if** $\frac{\sigma^L(B_L)}{\sigma^U(B^o_L)} \geq 1 - 1/e - \epsilon$ ***or*** $i = i_{\max}$ **then**
21 **return** B_L;
22 Double the sizes of $\mathbb{C}^s_1$ and $\mathbb{C}^s_2$ with new random CP sequences;

Algorithm 6: Max-Coverage($\mathbb{C}^s$, k)

1 $B \leftarrow \emptyset$;
2 **for** $i \leftarrow 1$ *to* k **do**
3 $u \leftarrow \arg\max_{v\in V\backslash S}(Cov_{\mathbb{C}^s}(B \cup \{v\}) - Cov_{\mathbb{C}^s}(B))$;
4 $B \leftarrow B \cup \{u\}$;
5 **return** B;

on the RIS framework [5], where the corresponding unbiased estimator (see Eq. (3)) is different in form from ours. As a result, the concentration bounds leveraged by OPIM-C are not tailored to our setting, thereby rendering its stopping criterion inapplicable. In this section, we first propose two new concentration bounds (Lemma 7), based on which we derive a novel stopping criterion. By integrating this criterion into the OPIM-C framework, we propose App-LB, an efficient algorithm that provides a $(1-1/e-\epsilon)$-approximate solution to the maximization of $D^L(\cdot)$ with at least $1-\delta$ probability.

Concentration bounds Given a set $\mathbb{C}^s$ of random CP sequences and a blocker set B, by defining a new martingale sequence, we derive concentration results tailored to describe how close between $Cov_{\mathbb{C}^s}(B)/|\mathbb{C}^s|$ and $D^L(B)$. Note that we assume $Cov_{C^s}(B) \leq \mathbb{E}[I_G(s)]$ for each $C^s \in \mathbb{C}^s$ as a necessary boundary condition for the derivation of our results. The condition is generally satisfied, given that $\mathbb{E}[\mathrm{Cov}_{C^s}(B)] = D^L(B) \ll \mathbb{E}[I_G(s)]$ holds under realistic budget settings.

Lemma 7 *Given a blocker set B, a seed node s and a set of θ random CP sequences $\mathbb{C}^s$. For any $\lambda > 0$,*

$$\Pr\left[\frac{Cov_{\mathbb{C}^s}(B)}{\mathbb{E}[I_G(s)]} - \frac{D^L(B)\cdot\theta}{\mathbb{E}[I_G(s)]} \geq \lambda\right] \leq \exp\left(-\frac{\lambda^2}{\frac{2D^L(B)}{\mathbb{E}[I_G(s)]}\cdot\theta + \frac{2}{3}\lambda}\right), \tag{9}$$

$$\Pr[\frac{Cov_{\mathbb{C}^s}(B)}{\mathbb{E}[I_G(s)]} - \frac{D^L(B)\cdot\theta}{\mathbb{E}[I_G(s)]} \leq -\lambda] \leq \exp\left(-\frac{\lambda^2\cdot\mathbb{E}[I_G(s)]}{2D^L(B)\cdot\theta}\right). \tag{10}$$

App-LB algorithm The pseudocode of App-LB is shown in Algorithm 5. Let ON be the set of outgoing neighbors of S (Line 1). When the budget k is no less than the number of nodes in ON, App-LB directly returns ON as the blocker set (Lines 2-3). Then we calculate $\hat{I}_G(s)$ with $(\gamma, \frac{\delta}{6})$-approximation by the stopping rule algorithm [65]. In addition, we derive the lower bound of $D^L(B_L^o)$ and define the constants $\theta_{\max}$ and θ_0 (Lines 4-7). Afterwards, we generate two sets of CP sequences $\mathbb{C}_1^s$ and $\mathbb{C}_2^s$, each of size θ_0, by repeatedly invoking CP-Gen procedure on G (Line 9). The subsequent part of the algorithm consists of at most i_{max} iterations. In each iteration, we first invoke Max-Coverage (Algorithm 6) to get a blocker set B_L, i.e., finding a set of k nodes such that B_L intersects with as many CP sequences as possible in $\mathbb{C}_1^s$ (Line 12). Then we derive $\sigma^L(B_L)$ and $\sigma^U(B_L^o)$ from $\mathbb{C}_2^s$ and $\mathbb{C}_1^s$, respectively (Lines 14-19). Specifically, $\sigma^L(B_L)$ is the lower bound of $D^L(B_L)/\mathbb{E}[I_G(s)]$ and $\sigma^U(B_L^o)$ is the upper bound of $D^L(B_L^o)/\mathbb{E}[I_G(s)]$. If $\sigma^L(B_L)/\sigma^U(B_L^o) \geq 1-1/e-\epsilon$ or $i = i_{\max}$, App-LB returns B_L and terminates. Otherwise, the quantities of CP sequences in $\mathbb{C}_1^s$ and $\mathbb{C}_2^s$ will be doubled, and App-LB will proceed to the next iteration (Lines 20-22). Next, we will explain in detail how App-LB can return a solution with approximation guarantees. Note that, when deriving $\theta_{\max}$, $\sigma^L(B_L)$ and $\sigma^U(B_L^o)$, it is imperative to carefully consider the error associated with estimating $\mathbb{E}[I_G(s)]$, to ensure the desired approximation guarantee.

Deriving $\theta_{\max}$. Based on previous concentration bounds [46], for the influence maximization problem, Tang et al. derive an upper bound on the sample size required to ensure $(1-1/e-\epsilon)$-approximation holds with probability at least $1-\delta$. Similarly, we derive the corresponding upper bound on the sample size for the IMIN problem based on our newly proposed concentration bounds. The following lemma provides the setting of θ_{max}, ensuring the correctness of App-LB when $i = i_{\max}$.

Lemma 8 *Let $\mathbb{C}^s$ be a set of random CP sequences, B_L be a size-k blocker set generated by applying Max-Coverage on $\mathbb{C}^s$, B_L^o be the optimal solution with size-k, OPT^L be the lower bound of $D^L(B_L^o)$ and $\hat{I}_G(s)$ be the estimated value of $\mathbb{E}[I_G(s)]$ with $(\gamma, \frac{\delta}{6})$-approximation. For fixed γ, ϵ and δ, let*

$$\theta_{max} = \frac{2\hat{I}_G(s)\left((1-1/e)\sqrt{\ln\frac{12}{\delta}} + \sqrt{(1-1/e)(\ln\binom{n-|S|}{k} + \ln\frac{12}{\delta})}\right)^2}{(1-\gamma)\epsilon^2 OPT^L},$$

if $|\mathbb{C}^s| = \theta \geq \theta_{\max}$, then B_L is $(1-1/e-\epsilon)$-approximate solution with at least $1-\delta/3$ probability.

Deriving OPT^L. Let $\Pr[s \to v]$ be the probability that s can activate v, and B^* be the set of k nodes in ON with the k largest probability of being activated by s. We set $\mathrm{OPT}^L = \sum_{v\in B^*}\Pr[s \to v]$. It can be verified that $\mathrm{OPT}^L \leq D^L(B_L^o)$, since for any $v \in V\backslash S$, $D^L(B\cup\{v\}) - D^L(B) \geq \Pr[s \to v]$ always holds. Moreover, the reason for selecting ON as the candidate set is that the value of $\Pr[s \to v]$ for each node $v \in ON$ can be computed efficiently, otherwise it will become the bottleneck of the whole algorithm.

Deriving $\sigma^L(B_L)$ *and* $\sigma^U(B_L^o)$. Subsequently, we show that $\sigma^L(B_L)$ (resp. $\sigma^U(B_L^o)$), as defined in Algorithm 5, is a lower bound of $D^L(B_L)/\mathbb{E}[I_G(s)]$ (resp. an upper bound of $D^L(B_L^o)/\mathbb{E}[I_G(s)]$) with a high probability.

Lemma 9 *For any $0 \leq \gamma, \epsilon, \delta \leq 1$, we have*

$$\Pr\left[\sigma^L(B_L) \leq D^L(B_L)/\mathbb{E}[I_G(s)]\right] \geq 1-\delta/3i_{\max},$$
$$\Pr\left[\sigma^U(B_L^o) \geq D^L(B_L^o)/\mathbb{E}[I_G(s)]\right] \geq 1-\delta/3i_{\max}.$$

Putting together The above results are sufficient to establish the approximation guarantees of App-LB. In particular, the algorithm has at most $i_{\max}$ iterations. In each of the first $i_{\max}-1$ iterations, by Lemma 9, $\frac{D^L(B_L)}{D^L(B_L^o)} \geq \frac{\sigma^L(B_L)}{\sigma^U(B_L^o)}$ holds with at least $1-2/3i_{\max}$ probability. This ensures that, once $\frac{\sigma^L(B_L)}{\sigma^U(B_L^o)}$ exceeds $1-1/e-\epsilon$, B_L achieves a $(1-1/e-\epsilon)$-approximation with at least $1-2\delta/3i_{\max}$ probability. Meanwhile, in the last iteration, a blocker set B_L is obtained under $|\mathbb{C}_1^s| \geq \theta_{\max}$; by Lemma 8, it follows that B_L is a $(1-1/e-\epsilon)$-approximate solution with at least $1-\delta/3$ probability. Using union bound, the following theorem holds.

Theorem 6 *Given* $0 \leq \gamma, \epsilon, \delta \leq 1$, *App-LB returns* B_L *satisfies* $\Pr[D^L(B_L) \geq (1 - 1/e - \epsilon)D^L(B_L^o)] \geq 1 - \delta$.

Moreover, Theorem 7 shows App-LB's time complexity.

Theorem 7 *Let* $\alpha(m, n)$ *be the inverse function of Ackerman's function with parameters* m *and* n. *When* $\frac{2\gamma}{1+\gamma} \leq \epsilon$ *and* $\delta \leq 1/2$, *the expected time complexity of App-LB algorithm is*

$$\mathcal{O}\left(\frac{(k \ln n + \ln 1/\delta)\mathbb{E}[I_G(s)](\alpha(m,n)m + \mathbb{E}[I_G(s)]^2)}{(\epsilon + \epsilon\gamma - 2\gamma)^2 D^L(B_L^o)} + \frac{m \cdot \ln 1/\delta}{\gamma^2}\right).$$

3.3.2 Upper bounding function maximization

This section then propose App-UB, an efficient algorithm that provides a $(1 - 1/e - \epsilon)$-approximate solution to the maximization of $D^U(\cdot)$ with at least $1 - \delta$ probability. From an algorithmic perspective, App-UB follows a similar framework to that of App-LB, with two key differences. First, two independent sets of random LRR sets, $\mathbb{L}_1$ and $\mathbb{L}_2$, are generated. Second, it adopts a different stopping condition for sample generation, as illustrated below.

$$\theta_{\max} = \frac{2|V_s'|\left((1-1/e)\sqrt{\ln\frac{6}{\delta}} + \sqrt{(1-1/e)(\ln\binom{|V_s'|-|S|}{k} + \ln\frac{6}{\delta})}\right)^2}{\epsilon^2 \mathrm{OPT}^L},$$

$$\theta_0 = \theta_{\max} \cdot \epsilon^2 \mathrm{OPT}^L / |V_s'|,$$

$$\sigma^L(B_U) = \left((\sqrt{Cov_{\mathbb{L}_2}(B_U) + \tfrac{2a_1}{9}} - \sqrt{\tfrac{a_1}{2}})^2 - \tfrac{a_1}{18}\right) \cdot \frac{|V_s'|}{|\mathbb{L}_2|},$$

$$\sigma^U(B_U^o) \leftarrow \left(\sqrt{Cov^u_{\mathbb{L}_1}(B_U^o) + \tfrac{a_2}{2}} + \sqrt{\tfrac{a_2}{2}}\right)^2 \cdot \frac{|V_s'|}{|\mathbb{L}_1|}.$$

To establish the approximation guarantees of App-UB, we introduce concentration bounds that characterize the deviation between $|V_s'| \cdot Cov_{\mathbb{L}}(B)/|\mathbb{L}|$ and $D^U(B)$, as presented in Lemma 10.

Lemma 10 *Given a blocker set* B, *a seed node* s *and a fixed number of* θ *random LRR sets* $\mathbb{L}$. *For any* $\lambda > 0$,

$$\Pr[Cov_{\mathbb{L}}(B) - \frac{D^U(B)\cdot\theta}{|V_s'|} \geq \lambda] \leq \exp(-\frac{\lambda^2}{\frac{2D^U(B)}{|V_s'|}\cdot\theta + \frac{2}{3}\lambda}), \quad (11)$$

$$\Pr[Cov_{\mathbb{L}}(B) - \frac{D^U(B)\cdot\theta}{|V_s'|} \leq -\lambda] \leq \exp(-\frac{|V_s'|\lambda^2}{2D^U(B)\cdot\theta}). \quad (12)$$

Leveraging Lemma 10, and following similar reasoning to that in Lemma 8 and Lemma 9, we can identify that in App-UB, i) $\theta_{\max}$ enables the blocker set B_U achieves the $(1 - 1/e - \epsilon)$-approximation in the worst case, and ii) $\frac{D^U(B_U)}{D^U(B_U^o)} \geq \frac{\sigma^L(B_U)}{\sigma^U(B_U^o)}$ holds with at least $1 - 2/3i_{\max}$ probability, in each of the first $i_{\max} - 1$ iterations. These results lead to the following theorem.

Theorem 8 *Given* $0 \leq \epsilon, \delta \leq 1$, *App-UB returns* B_U *satisfies* $\Pr[D^U(B_U) \geq (1 - 1/e - \epsilon)D^U(B_U^o)] \geq 1 - \delta$.

Moreover, Theorem 9 shows App-UB's time complexity.

Theorem 9 *When* $\delta \leq 1/2$, *the expected time complexity of App-UB is* $\mathcal{O}(\frac{(k \ln n + \ln 1/\delta)(m+n)}{\epsilon^2})$.

3.3.3 A lightweight heuristic for IMIN

This section further proposes a lightweight and simple heuristic, Heur-IMIN, for the IMIN problem. Heur-IMIN selects the blocker in a greedy manner from ON (the outgoing neighbors of S), guided by the quality function $s(\cdot)$ defined below, where $deg[v]$ is the out-degree of v.

$$s(v) = \Pr[s \rightarrow v] \cdot deg[v].$$

In other words, the k nodes with the largest $s(\cdot)$ values are selected as the blockers. Heur-IMIN is motivated by the following observations: i) the outgoing neighbors of the seeds are more likely to be blockers; ii) the nodes with a relatively high probability of being activated by s are more likely to be blockers; iii) the nodes with large influence are more likely to be blockers. If the budget is no less than the number of nodes in ON, we directly return ON as the blocker set. As indicated by Theorem 5, Heur-IMIN does not make any theoretical contribution to our final algorithm SandIMIN. Nevertheless, it may improve the empirical performance of SandIMIN with almost no extra runtime cost.

4 Solutions for blocker minimization

This section focuses on addressing the BM problem, for which we develop three algorithms. We first present a heuristic that ensures the threshold condition is satisfied (Sect. 4.1). We then design two approximation algorithms with non-trivial theoretical analysis, striking a trade-off: one achieves a tight approximation ratio at the expense of efficiency (Sect. 4.2), while the other provides a slightly looser guarantee with improved efficiency (Sect. 4.3). Notably, the approximation ratio here is defined in terms of the blocker set cardinality.

4.1 Heur-BM algorithm

Given the NP-hardness of the BM problem and #P-hardness of computing $D(\cdot)$, we first introduce the Heur-BM approach, an instantiation of the Thre-Greedy framework (Algorithm 1) that aims to optimize the unbiased estimator $\hat{D}(\cdot)$ of $D(\cdot)$. The basic idea of Heur-BM resembles that of the existing BM solution Base (see Sect. 2.4.2 for details). However, our Heur-BM approach theoretically guarantees that the threshold condition is satisfied with a high probability, owing to the unbiased nature of $\hat{D}(\cdot)$, while preserving efficiency through advanced algorithmic design.

At a high level, Heur-BM consists of two stages:

(1) **Blocker Selection.** This stage iteratively selects u with the largest $\hat{D}(u|B)$ and adds it to B, until the predefined stopping criterion is satisfied.
(2) **Verification.** This stage verifies whether the threshold condition (i.e., $D(B) \geq \eta$) is satisfied. If so, it returns B; otherwise, it continues selecting blockers until the condition is met.

In the first stage of Heur-BM, $\hat{D}(\cdot|B)$ is computed by a set of random CP sequences. Specifically, recall that the proposed lower bounding function $D^L(\cdot)$ only ignores the combination effect of the blockers in B, thus $D(v) = D^L(v)$ holds for any singleton node v. Accordingly, the estimator established for $D^L(v)$ (Lemma 5) also serves as an unbiased estimator for $D(v)$. That is, given a set of random CP sequences $\mathbb{C}^s$ generated on G, for each $v \in V\backslash S$, we have $D(v) = \mathbb{E}[Cov_{\mathbb{C}^s}(v)/|\mathbb{C}^s|]$. On this basis, let $\mathbb{C}^s_{V\backslash B}$ be a set of random CP sequences generated on induced subgraph $G[V\backslash B]$, for each $v \in V\backslash B$, it follows that $\hat{D}(v|B) = Cov_{\mathbb{C}^s_{V\backslash B}}(v)/|\mathbb{C}^s_{V\backslash B}|$ is an unbiased estimator for $D(v|B)$.

The second stage of Heur-BM utilizes the RIS approach and stopping rule (introduced in Sect. 2.5), to check whether the threshold condition is satisfied. In particular, it maintains two additional collections of random RR sets, denoted by $\mathcal{R}_1$ and $\mathcal{R}_2$. All the RR sets in $\mathcal{R}_1$ (resp. $\mathcal{R}_2$) are generated under G (resp. $G[V\backslash B]$). By ensuring $Cov_{\mathcal{R}_1}(S)$ and $Cov_{\mathcal{R}_2}(S)$ both exceed $\mathcal{T}(0, 1, \gamma, \delta)$ (Eq. (4)), we have $n \cdot Cov_{\mathcal{R}_1}(S)/|\mathcal{R}_1|$ and $n \cdot Cov_{\mathcal{R}_2}(S)/|\mathcal{R}_2|$ are (γ, δ)-estimates of $\mathbb{E}[I_G(S)]$ and $\mathbb{E}[I_{G[V\backslash B]}(S)]$, respectively [65], i.e.,

$$\Pr\left[n \cdot Cov_{\mathcal{R}_1}(S)/|\mathcal{R}_1| \leq (1+\gamma)\mathbb{E}[I_G(S)]\right] \geq 1 - \delta/2,$$
$$\Pr\left[n \cdot Cov_{\mathcal{R}_2}(S)/|\mathcal{R}_2| \geq (1-\gamma)\mathbb{E}[I_{G[V\backslash B]}(S)]\right] \geq 1 - \delta/2.$$

By the union bound, it follows that

$$D(B) = \mathbb{E}[I_G(S)] - \mathbb{E}[I_{G[V\backslash B]}(S)] \geq \frac{n \cdot Cov_{\mathcal{R}_1}(S)}{(1+\gamma)|\mathcal{R}_1|} - \frac{n \cdot Cov_{\mathcal{R}_2}(S)}{(1-\gamma)|\mathcal{R}_2|}.$$

holds with at least $1 - \delta$ probability. Therefore, once $\frac{n \cdot Cov_{\mathcal{R}_1}(S)}{(1+\gamma)|\mathcal{R}_1|} - \frac{n \cdot Cov_{\mathcal{R}_2}(S)}{(1-\gamma)|\mathcal{R}_2|}$ is not less than η, $D(B) \geq \eta$ holds with probability at least $1 - \delta$.

The pseudocode of Heur-BM is outlined in Algorithm 7. Initially, we generate θ_h random CP sequences on G and store them into $\mathbb{C}^s_{V\backslash B}$ (Lines 3-4). Then, in a similar manner to Thre-Greedy, we identify the blocker with the largest $Cov_{\mathbb{C}^s_{V\backslash B}}(\cdot)$ in each iteration (Lines 5-8). One difference is that, after selecting each blocker, we invoke Update-CP (Algorithm 8) to preserve the estimation accuracy for $D(\cdot|B)$. Specifically, for each CP sequence that intersects with B, we remove it and, as compensation, generate a new random CP sequence on $G[V\backslash B]$, ensuring that all sequences in $\mathbb{C}^s_{V\backslash B}$ exclude nodes in B. The blocker selection process terminates once $\Lambda/\theta_h \geq \eta$ is satisfied; the motivation for adopting this stopping condition will be clarified shortly.

Algorithm 7: Heur-BM(G, S, η, γ, δ)

Input : The graph G, the seed set S, the threshold η, the parameters γ, δ, the sample size θ_h.
Output : The blocker set B.

1 $\mathbb{C}^s_{V\backslash B} \leftarrow \emptyset$, $\mathcal{R}_1 \leftarrow \emptyset$, $\mathcal{R}_2 \leftarrow \emptyset$, $\Lambda \leftarrow 0$;
2 $\mathcal{T}(0, 1, \gamma, \delta) \leftarrow$ Eq. (4);
// First Stage: Blocker Selection
3 **while** $|\mathbb{C}^s_{V\backslash B}| < \theta_h$ **do**
4 $\quad C^s \leftarrow$ CP-Gen(G, S); $\mathbb{C}^s_{V\backslash B} \leftarrow \mathbb{C}^s_{V\backslash B} \cup \{C^s\}$;
5 **while** $\Lambda/\theta_h < \eta$ **do**
6 $\quad u \leftarrow \arg\max_{v \in V\backslash B} Cov_{\mathbb{C}^s_{V\backslash B}}(v)$;
7 $\quad B \leftarrow B \cup \{u\}$, $\Lambda \leftarrow \Lambda + Cov_{\mathbb{C}^s_{V\backslash B}}(u)$;
8 $\quad$ Update-CP($\mathbb{C}^s_{V\backslash B}$, B, S);
// Second Stage: Verification
9 **while** $Cov_{\mathcal{R}_1}(S) < \mathcal{T}(0, 1, \gamma, \delta)$ **do**
10 $\quad R \leftarrow$ RR-Gen(G); $\mathcal{R}_1 \leftarrow \mathcal{R}_1 \cup \{R\}$;
11 **while** $Cov_{\mathcal{R}_2}(S) < \mathcal{T}(0, 1, \gamma, \delta)$ **do**
12 $\quad R \leftarrow$ RR-Gen($G[V\backslash B]$); $\mathcal{R}_2 \leftarrow \mathcal{R}_2 \cup \{R\}$;
13 **while** $\frac{n \cdot Cov_{\mathcal{R}_1}(S)}{(1+\gamma)|\mathcal{R}_1|} - \frac{n \cdot Cov_{\mathcal{R}_2}(S)}{(1-\gamma)|\mathcal{R}_2|} < \eta$ **do**
14 $\quad u \leftarrow \arg\max_{v \in V\backslash B} Cov_{\mathbb{C}^s_{V\backslash B}}(v)$;
15 $\quad B \leftarrow B \cup \{u\}$;
16 $\quad$ Update-CP($\mathbb{C}^s_{V\backslash B}$, B, S);
17 $\quad$ Update-RR($\mathcal{R}_2$, B, $\mathcal{T}(0, 1, \gamma, \delta)$);
18 **return** B;

Algorithm 8: Update-CP($\mathbb{C}^s_{V\backslash B}$, B, S)

1 $cnt \leftarrow 0$;
2 **for** $C^s \in \mathbb{C}^s_{V\backslash B}$ **do**
3 $\quad$ **if** $B \cap C^s \neq \emptyset$ **then**
4 $\quad\quad \mathbb{C}^s_{V\backslash B} \leftarrow \mathbb{C}^s_{V\backslash B}\backslash\{C^s\}$; $cnt \leftarrow cnt + 1$;
5 **for** $i \leftarrow 1$ *to* cnt **do**
6 $\quad C^s \leftarrow$ CP-Gen($G[V\backslash B]$, S); $\mathbb{C}^s_{V\backslash B} \leftarrow \mathbb{C}^s_{V\backslash B} \cup \{C^s\}$;

Algorithm 9: Update-RR($\mathcal{R}_2$, B, $\mathcal{T}(0, 1, \gamma, \delta)$)

1 **for** $R \in \mathcal{R}_2$ **do**
2 $\quad$ **if** $B \cap R \neq \emptyset$ **then** $\mathcal{R}_2 \leftarrow \mathcal{R}_2\backslash\{R\}$;
3 **while** $Cov_{\mathcal{R}_2}(S) < \mathcal{T}(0, 1, \gamma, \delta)$ **do**
4 $\quad R \leftarrow$ RR-Gen($G[V\backslash B]$); $\mathcal{R}_2 \leftarrow \mathcal{R}_2 \cup \{R\}$;

Next, we repeatedly generating RR sets on G and $G[V\backslash B]$, which are stored in $\mathcal{R}_1$ and $\mathcal{R}_2$ respectively, until

the coverage of S in both exceeds $\mathcal{T}(0, 1, \gamma, \delta)$ (Lines 9-12). Next, if the condition in Line 13 is satisfied, which implies that $D(B) \geq \eta$ holds with high probability, the blocker set B is returned. Otherwise, we proceed to select the blocker also guided by $Cov_{\mathbb{C}^s_{V\setminus B}}(\cdot)$, while invoking both Update-CP and Update-RR, until the stopping condition is satisfied (Lines 13-17). Update-RR (Algorithm 9) is to preserve the estimation accuracy for $\mathbb{E}[I_{G[V\setminus B]}(S)]$ by updating $\mathcal{R}_2$: it removes RR sets intersecting with B and continues generating new ones on $G[V\setminus B]$ until $Cov_{\mathcal{R}_2}(S) \geq \mathcal{T}(0, 1, \gamma, \delta)$.

As observed, both stages of Heur-BM involve selecting blockers, but the second stage additionally requires executing Update-RR, which incurs extra computational cost. This motivates the introduction of the first stopping condition in Line 5 (i.e., $\Lambda/\theta_h \geq \eta$). Specifically, Λ/θ_h can be regarded as a rough (biased) estimate of $D(B)$, as illustrated by the following expression.

$$D(B) = \sum_{i=1}^{|B|} D(u_i^h | B_{i-1}^h) \approx \sum_{i=1}^{|B|} \hat{D}(u_i^h | B_{i-1}^h) = \sum_{i=1}^{|B|} Cov_{\mathbb{C}^s_{V\setminus B_{i-1}^h}}(u_i^h)/\theta_h = \Lambda/\theta_h.$$

where u_i^h is the i-th blocker selected by Heur-BM, and $B_i^h = \{u_1^h, u_2^h, \ldots, u_i^h\}$ is the set of the first i selected blockers. In the experiments, we empirically set $\theta_h = 10000$, which suffices to ensure the above approximate equality holds in most cases. As a result, once $\Lambda/\theta_h \geq \eta$ holds, $D(B)$ is likely to be close to η. This prevents the selection of excessive blockers, while reducing the number of iterations in the second stage, i.e., the number of Update-RR invocations, thereby enhancing overall efficiency.

Theoretical analysis As discussed previously, with at least $1-\delta$ probability, $\frac{n\cdot Cov_{\mathcal{R}_1}(s)}{(1+\gamma)|\mathcal{R}_1|} - \frac{n\cdot Cov_{\mathcal{R}_2}(s)}{(1-\gamma)|\mathcal{R}_2|} \leq D(B)$ holds. This leads to the following theorem.

Theorem 10 *Heur-BM returns B such that $D(B) \geq \eta$ holds with at least $1-\delta$ probability.*

Besides, Heur-BM's time complexity is shown below.

Theorem 11 *Let $\alpha(m, n)$ be the inverse function of Ackerman's function with parameters m and n, and $v^m = \arg\max_{v\in V} \mathbb{E}[I_G(v)]$. The expected time complexity of Heur-BM is*

$$\mathcal{O}\left(|V_s'|\left(\theta_h(m\cdot\alpha(m,n) + \mathbb{E}[I_G(s)]^2) + \frac{\ln\frac{1}{\delta} m\mathbb{E}[I_G(v^m)]}{\gamma^2|S|}\right)\right).$$

4.2 App-BM algorithm

While Heur-BM satisfies the threshold condition (Theorem 10), the non-submodularity of $D(\cdot)$ can lead any greedy-based method, including Heur-BM, to return a blocker set arbitrarily larger than the optimum.

Lemma 11 *For any method that greedily blocks the node of largest marginal gain w.r.t. $D(\cdot)$ until the threshold condition is met, the ratio $|B|/|B_2^o|$ for its returned blocker set B is unbounded, even when the objective is computed exactly.*

Proof Consider the following instance, in which all edge probabilities equal 1, so that all marginal gains are exact. The seed s has edges to nodes a and b, each of which has edges to all of $r_1, \ldots, r_M$; in addition, s has edges to $w_1, \ldots, w_L$, and each w_i has an edge to a further node w_i'. We set $\eta = M$ with $M \geq 2L + 2$.

We first determine $|B_2^o|$. Blocking a or b alone protects only that node, as every r_j stays reachable through the other, so $D(\{a\}) = D(\{b\}) = 1$; blocking both disconnects every r_j from s, giving $D(\{a, b\}) = M + 2$. Since no single node protects more than two nodes (w_i protects w_i and w_i') while $\eta = M > 2$, the threshold cannot be met by a single blocker but is met by $\{a, b\}$. It then follows that $|B_2^o| = 2$.

The greedy method first blocks $w_1, \ldots, w_L$, as each w_i has the maximum marginal gain 2 while every other node has gain 1, reaching $D = 2L$. It then continues until $D(B) \geq \eta = M$. The remaining deficit is $\eta - 2L = M - 2L \geq 2$, yet every still-available node gains at most 1 when blocked; the next pick therefore raises D to at most $2L+1 < \eta$, so at least two further blockers are required. Hence $|B| \geq L + 2$, and $|B|/|B_2^o| \geq (L + 2)/2 \to \infty$ as $L \to \infty$. This completes the proof. □

To address this limitation, this section designs App-BM, a bicriteria approximation algorithm for the BM problem. The main idea of App-BM is to optimize the submodular upper bound $D^U(\cdot)$ using the Thre-Greedy framework (Algorithm 1). Note that while this idea appears similar to that of the sandwich strategy, a key difference is that our goal here is to minimize the cardinality of the selected blocker set, whereas the sandwich strategy is designed for maximization problems. This intrinsically renders its theoretical results inapplicable.

As shown in Algorithm 10, App-BM first generates a certain number of LRR sets to compute the unbiased estimator $\hat{D}^U(\cdot)$ of $D^U(\cdot)$, according to Lemma 6. Then, Thre-Greedy is invoked to maximize $\hat{D}^U(\cdot)$ until $\hat{D}^U(B) \geq (1+\mu)\eta$, where μ is a input error parameter.

Theoretical analysis App-BM is much simpler than Heur-BM, but establishing its approximation guarantees is far from trivial. Let B_2^o (resp. B_2^*) be the solution with the smallest cardinality that satisfies $D(B_2^o) \geq \eta$ (resp. $D^U(B_2^*) \geq \eta$). Besides, let u_i^a be the i-th blocker selected by App-BM, and $B_i^a = \{u_1^a, u_2^a, \ldots, u_i^a\}$ be the blockers selected in the first i iterations. At a high level, we bound the cardinality of App-BM's solution using the submodular-cover guarantee

Algorithm 10: App-BM(G, S, η, μ, δ)

Input : The graph G, the seed set S, the threshold η, and the error parameters μ, δ.
Output : The blocker set B, the sample collection $\mathbb{L}$.

$\mathbb{L} \leftarrow \emptyset$;
$\theta_a \leftarrow \frac{(\frac{2}{3}\mu+2)|V_s'|\ln\left(\frac{3(\lceil\eta\rceil-1)}{\delta}\cdot\left(\frac{en}{\lceil\eta\rceil-1}\right)^{\lceil\eta\rceil-1}\right)}{\eta\mu^2}$;
while $|\mathbb{L}| < \theta_a$ **do**
  $L \leftarrow$ LRR-Gen(G, S); $\mathbb{L} \leftarrow \mathbb{L} \cup \{L\}$;
$\hat{D}^U(\cdot) = |V_s'| \cdot Cov_{\mathbb{L}}(\cdot)/\theta_a$;
$B \leftarrow$ Thre-Greedy($G, (1+\mu)\eta, \hat{D}^U$);
return $(B, \mathbb{L})$;

(Theorem 4), which we adapt in two respects. First, that guarantee is relative to B_2^*, whereas we seek a bound relative to the true optimum B_2^o. Since $D^U \geq D$ makes the threshold easier to meet, $|B_2^*| \leq |B_2^o|$, and the bound carries over directly. Second, the guarantee assumes the exact D^U, whereas App-BM optimizes only its estimator $\hat{D}^U$. Accounting for this approximation is the crux of the analysis, which we handle through three lemmas (Lemmas 12–14) before assembling them into Theorem 12.

Intuitively, the algorithm stops at the slightly inflated target $(1+\mu)\eta$ rather than η, leaving a safety margin that absorbs the small estimation error. The first lemma then guarantees that, with enough samples, App-BM's solution reaches the threshold under D^U.

Lemma 12 *If $|\mathbb{L}| \geq \theta_a$, then with probability at least $1-\delta/3$, the solution B returned by Thre-Greedy$(G, (1+\mu)\eta, \hat{D}^U)$ satisfies $D^U(B) \geq \eta$.*

Proof We call B is bad if $D^U(B) < \eta$. To prove this lemma, we will show that each bad blocker set B' has at most $\delta/3\sum_{k_i=1}^{\lceil\eta\rceil-1}\binom{n}{k_i}$ probability to be returned by Thre-Greedy $(G, (1+\mu)\eta, \hat{D}^U)$.

This is sufficient to establish the lemma since there exists at most $\sum_{k_i=1}^{\lceil\eta\rceil-1}\binom{n}{k_i}$ bad blocker sets, and by the union bound, there is at least $1-\delta/3$ probability that none of them is returned.

Define $\lambda_1 = |\mathbb{L}| \cdot \frac{\mu\eta}{|V_s'|}$ and $k_m = \lceil\eta\rceil - 1$, we have

$$\begin{aligned}
&\Pr[\hat{D}^U(B') \geq (1+\mu)\eta] \leq \Pr[\hat{D}^U(B') - D^U(B') \geq \mu\eta] \\
&= \Pr[Cov_{\mathbb{L}}(B') - |\mathbb{L}| \cdot \frac{D^U(B')}{|V_s'|} \geq \lambda_1] \\
&\leq \exp\left(\frac{-\lambda_1^2}{\frac{2\lambda_1}{3} + \frac{2D^U(B')|\mathbb{L}|}{|V_s'|}}\right) \leq \frac{\delta}{3k_m \cdot (en/k_m)^{k_m}},
\end{aligned}$$

where the first inequality is due to $D^U(B') < \eta$, the second inequality is due to Eq. (11), the third inequality is due to $|\mathbb{L}| \geq \theta_a$ and $D^U(B') < \eta$. Also, we have

$$\sum_{k_i=1}^{k_m}\binom{n}{k_i} \leq \sum_{k_i=1}^{k_m}\frac{n^{k_i}}{k_i!} \leq k_m \cdot (en/k_m)^{k_m}.$$

where the second inequality is due to $k_i! \geq (\frac{k_i}{e})^{k_i}$.

On this basis, we have

$$\Pr[\hat{D}^U(B') \geq (1+\mu)\eta] \leq \frac{\delta}{3k_m(en/k_m)^{k_m}} \leq \frac{\delta}{3\sum_{k_i=1}^{k_m}\binom{n}{k_i}}.$$

This completes the proof. □

Moreover, since $\hat{D}^U$ is unbiased, sufficient samples make it accurate, which gives the following lemma.

Lemma 13 *Define*

$$\Psi(\eta, \mu, \delta) = 2|V_s'|\ln\frac{1}{\delta}/\eta\mu^2. \tag{13}$$

If $|\mathbb{L}| \geq \Psi(\eta, \mu, \delta/3)$, then we have $\hat{D}^U(B_2^*) \geq (1-\mu)D^U(B_2^*)$ holds with at least $1-\delta/3$ probability.

Proof Let $\lambda_2 = \frac{\mu|\mathbb{L}|D^U(B_2^*)}{|V_s'|}$, we have

$$\begin{aligned}
&\Pr[\hat{D}^U(B_2^*) \leq (1-\mu)D^U(B_2^*)] \\
&= \Pr[Cov_{\mathbb{L}}(B_2^*) - \frac{|\mathbb{L}|D^U(B_2^*)}{|V_s'|} \leq -\lambda_2] \\
&\leq \exp\left(-\frac{\lambda_2^2|V_s'|}{2D^U(B_2^*)\cdot|\mathbb{L}|}\right) \leq \exp\left(-\frac{\mu^2|\mathbb{L}|\eta}{2|V_s'|}\right) \leq \delta/3,
\end{aligned}$$

where the first inequality is due to Eq. (12), the second inequality is due to $D^U(B_2^*) \geq \eta$, and the third inequality is due to $|\mathbb{L}| \geq \Psi(\eta, \mu, \delta/3)$. □

It can be verified that $\theta_a > \Psi(\eta, \mu, \delta/3)$. Thus the results in Lemma 13 also hold when $|\mathbb{L}| \geq \theta_a$. Now let $B_q^a = B$ be the solution returned by App-BM (at $|\mathbb{L}| = \theta_a$), with $|B_q^a| = q$, and let B_p^a be the output of Thre-Greedy$(G, (1-\mu)\eta, \hat{D}^U)$ on the same LRR sets, with $|B_p^a| = p$. Intuitively, since the targets $(1+\mu)\eta$ and $(1-\mu)\eta$ differ only slightly, the greedy procedure selects nearly the same blockers in both runs. The following lemma formalizes this by bounding the difference between p and q.

Lemma 14 *For any $0 < \mu \leq \frac{\mathbb{E}[I_G(S)]-|S|-\eta}{\mathbb{E}[I_G(S)]-|S|+\eta+2|V_s'|}$, $q \leq p+1$ holds with at least $1-\delta/3$ probability when $|\mathbb{L}| = \theta_a$.*

Proof To prove this lemma, we consider two cases: *i)* $\hat{D}^U(B_p^a) \geq (1+\mu)\eta$; and *ii)* $\hat{D}^U(B_p^a) < (1+\mu)\eta$. For the first case, we have $p = q$. For the second case, since $|\mathbb{L}| = \theta_a$ and $D^U(V_s') = \mathbb{E}[I_G(S)] - |S| \geq \eta$, similar to

the reasoning in Lemma 13, we can derive that the following equation holds with at least $1-\delta/3$ probability.

$$\hat{D}^U(V_s') \geq (1-\mu)D^U(V_s'). \tag{14}$$

Assuming that Eq. (14) holds. For any i, we have

$$\begin{aligned}
&(1-\mu)(\mathbb{E}[I_G(S)]-|S|)-\hat{D}^U(B_i^a)\\
=&(1-\mu)D^U(V_s')-\hat{D}^U(B_i^a)\leq \hat{D}^U(V_s')-\hat{D}^U(B_i^a)\\
\leq&\sum_{v\in V_s'\setminus B_i^a}(\hat{D}^U(B_i^a\cup\{v\})-\hat{D}^U(B_i^a))\\
\leq&|V_s'|\cdot(\hat{D}^U(B_{i+1}^a)-\hat{D}^U(B_i^a)).
\end{aligned}\tag{15}$$

where the second inequality is due to the submodularity of $\hat{D}^U(\cdot)$. On this basis, we can further derive that

$$\begin{aligned}
&\hat{D}^U(B_{p+1}^a)-\hat{D}^U(B_p^a)\\
\geq&\frac{(1-\mu)(\mathbb{E}[I_G(S)]-|S|)-\hat{D}^U(B_p^a)}{|V_s'|}\\
\geq&\frac{(1-\mu)(\mathbb{E}[I_G(S)]-|S|)-(1+\mu)\eta}{|V_s'|}\geq 2\mu\eta,
\end{aligned}$$

where the second inequality is due to $\hat{D}^U(B_p^a)<(1+\mu)\eta$, and the third inequality is due to $\mu\leq\frac{\mathbb{E}[I_G(S)]-|S|-\eta}{\mathbb{E}[I_G(S)]-|S|+\eta+2|V_s'|}$. We then have

$$\begin{aligned}
\hat{D}^U(B_{p+1}^a)&\geq\hat{D}^U(B_p^a)+2\mu\eta\\
&\geq(1-\mu)\eta+2\mu\eta\geq(1+\mu)\eta.
\end{aligned}$$

Thus we have $p+1\geq q$. Recall that Eq. (14) holds with at least $1-\delta/3$ probability, the lemma holds. □

Main results for App-BM. Given a sample collection $\mathbb{L}$ and its associated estimator $\hat{D}^U(\cdot)$, let $B_{\mathbb{L}}^{\mu}$ be the minimum-cardinality solution that satisfies $\hat{D}^U(B_{\mathbb{L}}^{\mu})\geq(1-\mu)\eta$ and $\tau(B,\mathbb{L})=\max_{v\in V}\hat{D}^U(v|B_{|B|-1})$. Combining the three lemmas with the submodular-cover bound (Theorem 4) yields the main guarantee of App-BM.

Theorem 12 *For any* $0<\mu\leq\frac{\mathbb{E}[I_G(S)]-|S|-\eta}{\mathbb{E}[I_G(S)]-|S|+\eta+2|V_s'|}$ *and* $\delta\in(0,1)$, *App-BM returns* $(B,\mathbb{L})$ *such that*

$$D(B)\geq\frac{D(B)}{D^U(B)}\cdot\eta \text{ and } |B|\leq\min\{\tilde{\beta}_1,\tilde{\beta}_2\}\cdot|B_2^o|+1$$

holds with at least $1-\delta$ *probability, where*

$$\begin{aligned}
\tilde{\beta}_1&=1+\ln\tfrac{(1-\mu)\eta}{\tau(B,\mathbb{L})\cdot|B_{\mathbb{L}}^{\mu}|}\leq 1+\ln|\mathbb{L}|,\\
\tilde{\beta}_2&=\lceil\ln\tfrac{n\eta}{n-\eta}\rceil+\tfrac{1}{|B_2^o|}\leq 2\ln n+2.
\end{aligned}$$

Proof By applying the union bound to Lemmas 12, 13, and 14, we obtain that the following equations hold simultaneously with at least $1-\delta$ probability if $|\mathbb{L}|=\theta_a$.

$$D^U(B)\geq\eta \tag{16}$$

$$\hat{D}^U(B_2^*)\geq(1-\mu)D^U(B_2^*) \tag{17}$$

$$q\leq p+1. \tag{18}$$

Assuming all these equations hold. By Eq. (16), we have

$$D(B)=\frac{D(B)}{D^U(B)}\cdot D^U(B)\geq\frac{D(B)}{D^U(B)}\cdot\eta.$$

By Eq. (17), $\hat{D}^U(B_2^*)\geq(1-\mu)D^U(B_2^*)\geq(1-\mu)\eta$ holds. Then, it follows that $|B_{\mathbb{L}}^{\mu}|\leq|B_2^*|$. In addition, by the definition of B_2^o and B_2^*, it can verified that $|B_2^*|\leq|B_2^o|$. As a result, we have $|B_{\mathbb{L}}^{\mu}|\leq|B_2^o|$.

Recall that B_p^a is the solution returned by Thre-Greedy $(G,(1-\mu)\eta,\hat{D}^U)$, by Theorem 4, we have

$$\begin{aligned}
|B_p^a|&\leq(1+\ln\frac{(1-\mu)\eta}{\max_{v\in V}\hat{D}^U(v|B_{p-1}^a)\cdot|B_{\mathbb{L}}^{\mu}|})\cdot|B_{\mathbb{L}}^{\mu}|\\
&\leq(1+\ln\frac{(1-\mu)\eta}{\tau(B,\mathbb{L})\cdot|B_{\mathbb{L}}^{\mu}|})\cdot|B_{\mathbb{L}}^{\mu}|, \text{ and}\\
|B_p^a|&\leq\lceil\ln\frac{n\eta}{n-\eta}\rceil|B_{\mathbb{L}}^{\mu}|+1.
\end{aligned}$$

Combining the above two equations with Eq. (18) and $|B_{\mathbb{L}}^{\mu}|\leq|B_2^o|$, we have

$$\begin{aligned}
|B|=|B_q^a|\leq|B_p^a|+1&\leq(1+\ln\frac{(1-\mu)\eta}{\tau(B,\mathbb{L})|B_{\mathbb{L}}^{\mu}|})\cdot|B_{\mathbb{L}}^{\mu}|+1\\
&\leq\tilde{\beta}_1|B_2^o|+1,\\
\text{and } |B|\leq|B_p^a|+1&\leq\lceil\ln\frac{n\eta}{n-\eta}\rceil|B_{\mathbb{L}}^{\mu}|+2\\
&\leq\lceil\ln\frac{n\eta}{n-\eta}\rceil|B_2^o|+2=\tilde{\beta}_2|B_2^o|+1.
\end{aligned}$$

It remains to verify the upper bounds on $\tilde{\beta}_1$ and $\tilde{\beta}_2$. Let $g_{\max}=\max_{v\in V}\hat{D}^U(v)$ be the maximum single-node estimate on $\mathbb{L}$. For $\tilde{\beta}_1$: as $B_{\mathbb{L}}^{\mu}$ is a minimum-cardinality set with $\hat{D}^U(B_{\mathbb{L}}^{\mu})\geq(1-\mu)\eta$, the submodularity of the estimator $\hat{D}^U(\cdot)$ gives $(1-\mu)\eta\leq\hat{D}^U(B_{\mathbb{L}}^{\mu})\leq\sum_{v\in B_{\mathbb{L}}^{\mu}}\hat{D}^U(v)\leq|B_{\mathbb{L}}^{\mu}|\cdot g_{\max}$. Moreover, since $\hat{D}^U(\cdot)=|V_s'|\cdot Cov_{\mathbb{L}}(\cdot)/|\mathbb{L}|$ with $Cov_{\mathbb{L}}(\cdot)$ integer-valued, the positive last marginal gain satisfies $\tau(B,\mathbb{L})\geq|V_s'|/|\mathbb{L}|\geq g_{\max}/|\mathbb{L}|$. As a result, $\tilde{\beta}_1\leq 1+\ln\frac{g_{\max}}{\tau(B,\mathbb{L})}\leq 1+\ln|\mathbb{L}|$. For $\tilde{\beta}_2$: since $\eta<\mathbb{E}[I_G(S)]-|S|$ and $\mathbb{E}[I_G(S)]\leq n$, we have $n-\eta>|S|\geq 1$ and $\eta<n$, so $\lceil\ln\frac{n\eta}{n-\eta}\rceil\leq 2\ln n+1$; thus $\tilde{\beta}_2\leq 2\ln n+2$ by $|B_2^o|\geq 1$. □

The time complexity of App-BM is established below.

Theorem 13 *Let* $v^o = \arg\max_{v \in V} D^U(v)$. *The expected runtime of App-BM is* $\mathcal{O}(\frac{(\frac{1}{\eta}\ln\frac{\eta}{\delta}+\ln\frac{n}{\eta})mD^U(v^o)}{\mu})$.

4.3 App-BM+ algorithm

Algorithm 11: App-BM+(G, S, η, δ)

Input : The graph G, the seed set S, the threshold η, and the parameter δ.
Output : The blocker set B, the sample collection $\mathbb{L}_1$ and the parameter μ.

1 Initialize $B \leftarrow \emptyset$, $B_p^u \leftarrow \emptyset$, $B_q^u \leftarrow \emptyset$, $\mathbb{L}_1 \leftarrow \emptyset$;
2 $\tilde{\mu} \leftarrow \frac{1}{3|V_s'|-|S|+\eta}$, $\mu \leftarrow 0.5$;
3 $\theta_{\max} \leftarrow \frac{(\frac{2}{3}\tilde{\mu}+2)|V_s'|\ln\left(\frac{9\lceil\eta\rceil-1)}{\delta}\cdot(\frac{en}{\lceil\eta\rceil-1})^{\lceil\eta\rceil-1}\right)}{\eta\tilde{\mu}^2}$;
4 $\Psi(\eta, \mu, \frac{\delta}{9}) \leftarrow$ Eq. (13); $\mathcal{T}(0, 1, \mu, \frac{2}{9}\delta) \leftarrow$ Eq. (4);
5 **while** *true* **do**
6 **while** $|\mathbb{L}_1| < \Psi(\eta, \mu, \frac{\delta}{9})$ **do**
7 $L \leftarrow$ LRR-Gen(G, S); $\mathbb{L}_1 \leftarrow \mathbb{L}_1 \cup \{L\}$;
8 $\hat{D}^U(\cdot) = |V_s'| \cdot Cov_{\mathbb{L}_1}(\cdot)/|\mathbb{L}_1|$;
9 **if** $|\mathbb{L}_1| \geq \theta_{\max}$ **then**
10 $B \leftarrow$ Thre-Greedy(G, $(1+\tilde{\mu})\eta$, $\hat{D}^U$);
11 **return** $(B, \mathbb{L}_1, \tilde{\mu})$;
12 $B_p^u \leftarrow$Thre-Greedy (G, $(1-\mu)\eta$, $\hat{D}^U$);
13 $B_q^u \leftarrow$Thre-Greedy (G, $(1+\mu)\eta$, $\hat{D}^U$);
 // Condition I
14 **if** $|B_q^u| \leq 2|B_p^u|$ **then**
15 $\mathbb{L}_2 \leftarrow \emptyset$;
16 **while** $Cov_{\mathbb{L}_2}(B_q^u) < \mathcal{T}(0, 1, \frac{\mu}{2}, \frac{2}{9}\delta)$ **do**
17 $L \leftarrow$ LRR-Gen(G, S); $\mathbb{L}_2 \leftarrow \mathbb{L}_2 \cup \{L\}$;
 // Condition II
18 **if** $|V_s'| \cdot Cov_{\mathbb{L}_2}(B_q^u)/|\mathbb{L}_2| \geq (1+\mu/2)\cdot\eta$ **then**
19 $B \leftarrow B_q^u$; **return** $(B, \mathbb{L}_1, \mu)$;
20 **else**
21 $\mu \leftarrow \mu/\sqrt{2}$;
22 $\mathbb{L}_1 \leftarrow \mathbb{L}_1 \cup \mathbb{L}_2$, $\delta \leftarrow \delta/2$;

As established by Theorem 12, App-BM provides a non-trivial bicriteria approximation guarantee for BM problem. Nevertheless, one notable drawback of App-BM is that the input estimation error μ is limited by an upper bound $\frac{\mathbb{E}[I_G(S)]-|S|-\eta}{\mathbb{E}[I_G(S)]-|S|+\eta+2|V_s'|}$, which can be very small due to the potentially large value of $|V_s'|$. Accordingly, as indicated by the setting of θ_a, a substantial number of random LRR sets would be required, making App-BM impractical and primarily of theoretical interest.

To resolve this issue, we design App-BM+, a more efficient bicriteria approximation approach that eliminates the restriction on μ. In a nutshell, App-BM+ also adopts Thre-Greedy (Algorithm 1) to optimize $\hat{D}^U(\cdot)$. The key difference lies in its use of a *trial-and-error* strategy to dynamically decrease μ until the pre-defined stopping conditions are satisfied. This adaptive mechanism results in a relatively large final μ, thereby ensuring a practical sample size.

As outlined in Algorithm 11, App-BM+ works iteratively, with μ initially set to 0.5. At the start of each iteration, a number of random LRR sets are generated (if needed) and stored in $\mathbb{L}_1$ (Lines 6-7). Guided by the estimator derived from $\mathbb{L}_1$, Thre-Greedy is then invoked twice to derive B_p^u and B_q^u, using thresholds $(1-\mu)\eta$ and $(1+\mu)\eta$, respectively (Lines 12-13). B_q^u is the candidate solution, while B_p^u is the auxiliary solution used to evaluate whether μ is sufficiently small, based on `Condition I` in Line 14. If the condition is satisfied, we further check whether $D^U(B_q^u) \geq \eta$ holds, based on `Condition II` (Lines 15–19). In particular, we first initialize an empty collection $\mathbb{L}_2$, and then repeatedly generate random LRR sets and store them into $\mathbb{L}_2$ until $Cov_{\mathbb{L}_2}(B_q^u) \geq \mathcal{T}(0, 1, \frac{\mu}{2}, \frac{2\delta}{9})$. Note that, the maintenance of two independent sample collections is essential, as reusing one for both identifying B_q^u and computing $\hat{D}^U(B_q^u)$ would introduce estimation bias. Based on the stopping rule for sampling [65] (see Sect. 2.5), the resulting estimate provides an $(\frac{\mu}{2}, \frac{2\delta}{9})$-approximation to $D^U(B_q^u)$. Hence, satisfying `Condition II` (Line 18) implies $D^U(B_q^u) \geq \eta$ holds with high probability; in this case, B_q^u is returned. In contrast, when `Condition I` fails to hold, the value of μ is reduced by a factor of $\sqrt{2}$ (Lines 20-21). Finally, the sample collection $\mathbb{L}_2$ is merged into $\mathbb{L}_1$ and δ is halved, aiming to find a better candidate solution in the next iteration (Line 22). The total number of samples is capped by $\theta_{\max}$ (initialized in Line 3); once $|\mathbb{L}_1|$ exceeds $\theta_{\max}$, the solution produced by Thre-Greedy is directly returned (Lines 9-11).

Theoretical analysis The approximation guarantee of App-BM+ follows from analyzing its two termination cases. The first, when the sample cap is reached, is handled by Lemma 15; the second, when both stopping conditions are met, by Lemma 18, which builds on Lemmas 16 and 17, establishing that the estimates remain accurate across all iterations. Combining the two cases yields the overall guarantee of App-BM+ (Theorem 14).

We begin with the first case, where App-BM+ reduces to App-BM with $\mu = \tilde{\mu}$ and thus inherits its guarantee from Theorem 12.

Lemma 15 *Let* $(B, \mathbb{L}_1, \tilde{\mu})$ *denote the tuple returned by App-BM+ when* $|\mathbb{L}_1| \geq \theta_{\max}$. *For any* $\eta \in (0, \mathbb{E}[I_G(S)]-|S|-1]$, *we have*

$$D(B) \geq \frac{D(B)}{D^U(B)} \cdot \eta \text{ and } |B| \leq \lceil \ln \tfrac{n\eta}{n-\eta} \rceil |B_2^o| + 2$$

holds with at least $1-\delta/3$ *probability.*

Proof It can be verified that $\tilde{\mu} \leq \frac{\mathbb{E}[I_G(S)]-|S|-\eta}{\mathbb{E}[I_G(S)]-|S|+\eta+2|V_s'|}$. Then, by Theorem 12, the lemma holds. □

To facilitate the analysis of the second case, we establish two supporting lemmas, both following from the unbiasedness of the estimators and the union bound.

Lemma 16 *For all iterations, with at least* $1-\frac{2\delta}{9}$ *probability,* $\hat{D}^U(B_2^*) \geq (1-\mu)D^U(B_2^*)$ *holds.*

Proof In the i-iteration of App-BM+, we have $|\mathbb{L}_1| \geq \Psi(\eta, \mu, \frac{\delta}{9\cdot 2^{i-1}})$. By Lemma 13, it follows that

$$\Pr[\hat{D}^U(B_2^*) \geq (1-\mu)D^U(B_2^*)] \geq 1-\delta/(9\cdot 2^{i-1}).$$

Let $i_{\max}$ be the maximum number of iterations performed by App-BM+. We have $1-\sum_{i=1}^{i_{\max}} \frac{\delta}{9\cdot 2^{i-1}} \geq 1-\frac{2\delta}{9}$. By the union bound, the lemma holds. □

Lemma 17 *For all iterations that Condition I is met,* $|V_s'|\cdot Cov_{\mathbb{L}_2}(B_q^u)/|\mathbb{L}_2| \leq (1+\mu/2)D^U(B_q^u)$ *holds with at least* $1-\frac{2\delta}{9}$ *probability.*

Proof In App-BM+'s i-iteration that `Condition I` is met, we have $Cov_{\mathbb{L}_2}(B_q^u) \geq \mathcal{T}(0,1,\frac{\mu}{2},\frac{2\delta}{9\cdot 2^{i-1}})$. By the stopping rule for sampling [65], we have

$$\Pr\left[\frac{|V_s'|Cov_{\mathbb{L}_2}(B_q^u)}{|\mathbb{L}_2|} \leq (1+\frac{\mu}{2})D^U(B_q^u)\right] \geq 1-\frac{\delta}{9\cdot 2^{i-1}}.$$

Similar to reasoning in Lemma 16, the lemma holds. □

Based on these, we establish the following guarantee for the second case. Intuitively, Condition II verifies the returned set on fresh samples to confirm that it clears the threshold, while Condition I caps its size at twice that of the lower-threshold greedy run, which is itself near-optimal by the submodular-cover guarantee.

Lemma 18 *Let* $(B, \mathbb{L}_1, \mu)$ *denote the tuple returned by App-BM+ when B satisfies both Condition I and Condition II. We have*

$$D(B) \geq \frac{D(B)}{D^U(B)}\cdot\eta \text{ and } |B| \leq 2\min\{\hat{\beta}_1, \hat{\beta}_2\}\cdot|B_2^o|$$

holds with at least $1-\frac{4\delta}{9}$ *probability, where* $\hat{\beta}_1 = 1+\ln\frac{(1-\mu)\eta}{\tau(B,\mathbb{L}_1)\cdot|B_{\mathbb{L}_1}^{\mu}|}$ *and* $\hat{\beta}_2 = \lceil\ln\frac{n\eta}{n-\eta}\rceil + \frac{1}{|B_2^o|}$.

Proof By Lemmas 16, 17 and the union bound, for all iterations that `Condition I` is met, the following equations hold with at least $1-\frac{4\delta}{9}$ probability.

$$\hat{D}^U(B_2^*) \geq (1-\mu)D^U(B_2^*), \tag{19}$$

$$|V_s'|\cdot Cov_{\mathbb{L}_2}(B_q^u)/|\mathbb{L}_2| \leq (1+\mu/2)D^U(B_q^u). \tag{20}$$

Assuming these equations hold simultaneously. By Eq. (20), when `Condition II` is satisfied, we have

$$D^U(B) = D^U(B_q^u) \geq \frac{|V_s'|\cdot Cov_{\mathbb{L}_2}(B_q^u)}{|\mathbb{L}_2|(1+\mu/2)} \geq \eta.$$

On this basis, we then have $D(B) \geq \frac{D(B)}{D^U(B)}\cdot\eta$. In addition, based on Eq. (19) and applying reasoning similar to that in Theorem 12, we can derive $|B_{\mathbb{L}_1}^{\mu}| \leq |B_2^o|$. Moreover, by Theorem 4, it follows that

$$|B_p^u| \leq (1+\ln\frac{(1-\mu)\eta}{\max_{v\in V}\hat{D}^U(v|B_{p-1}^u)\cdot|B_{\mathbb{L}_1}^{\mu}|})\cdot|B_{\mathbb{L}_1}^{\mu}|$$
$$\leq (1+\ln\frac{(1-\mu)\eta}{\tau(B,\mathbb{L}_1)\cdot|B_{\mathbb{L}_1}^{\mu}|})\cdot|B_{\mathbb{L}_1}^{\mu}|, \text{ and}$$
$$|B_p^u| \leq \lceil\ln\frac{n\eta}{n-\eta}\rceil\cdot|B_{\mathbb{L}_1}^{\mu}|+1$$

When `Condition I` is met, we have $|B_q^u| \leq 2|B_p^u|$. By combining all the above results, we have

$$|B| = |B_q^u| \leq 2|B_p^u| \leq 2(1+\ln\frac{(1-\mu)\eta}{\tau(B,\mathbb{L}_1)\cdot|B_{\mathbb{L}_1}^{\mu}|})|B_{\mathbb{L}_1}^{\mu}|$$
$$\leq 2(1+\ln\frac{(1-\mu)\eta}{\tau(B,\mathbb{L}_1)\cdot|B_{\mathbb{L}_1}^{\mu}|})|B_2^o|, \text{ and}$$
$$|B| \leq 2|B_p^u| \leq 2(\lceil\ln\frac{n\eta}{n-\eta}\rceil+\frac{1}{|B_2^o|})|B_2^o|$$

This completes the proof. □

Main results for App-BM+. Finally, by jointly considering the theoretical results of both cases, we establish the performance guarantees of App-BM+.

Theorem 14 *For any* $\eta \in (0, \mathbb{E}[I_G(S)]-|S|-1]$, *App-BM+ returns* $(B, \mathbb{L}_1, \mu)$ *such that*

$$D(B) \geq \frac{D(B)}{D^U(B)}\cdot\eta \text{ and } |B| \leq \max\{\min\{2\hat{\beta}_1, 2\hat{\beta}_2\}, \hat{\beta}_3\}|B_2^o|$$

holds with at least $1-\delta$ *probability, where*

$$\hat{\beta}_1 = 1+\ln\frac{(1-\mu)\eta}{\tau(B,\mathbb{L}_1)\cdot|B_{\mathbb{L}_1}^{\mu}|} \leq 1+\ln|\mathbb{L}_1|,$$
$$\hat{\beta}_2 = \lceil\ln\frac{n\eta}{n-\eta}\rceil + \frac{1}{|B_2^o|} \leq 2\ln n+2,$$
$$\hat{\beta}_3 = \lceil\ln\frac{n\eta}{n-\eta}\rceil + \frac{2}{|B_2^o|} \leq 2\ln n+3.$$

Proof The performance guarantee follows by applying the union bound over the results of Lemmas 15 and 18. The upper bounds on the three factors follow by the same arguments as in the proof of Theorem 12. □

In addition, the time complexity of App-BM+ is established by the following theorem.

Table 2 Statistics of datasets

Dataset	Type	$\|V\|$	$\|E\|$	Avg. deg
EmailCore (EC)	directed	1,005	25,571	49.6
Facebook (FB)	undirected	4,039	88,234	43.7
Wiki-Vote (WV)	directed	7,115	103,689	29.1
EmailAll (EA)	directed	265,214	420,045	3.2
DBLP (DB)	undirected	317,080	1,049,866	6.6
Twitter (TT)	directed	81,306	1,768,149	59.5
Stanford (SF)	directed	281,903	2,312,497	16.4
Youtube (YT)	undirected	1,134,890	2,987,624	5.3
Pokec (PK)	directed	1,632,803	30,622,564	37.5

Theorem 15 *Let* $v^o = \arg\max_{v\in V} D^U(v)$. *The expected time complexity of App-BM+ is*

$$\mathcal{O}\left(\frac{(\frac{1}{\eta}\ln\frac{\eta}{\delta}+\ln\frac{n}{\eta})^2|V_s'|^3\mathbb{E}[I_G(S)]D^U(v^o)}{\ln\frac{1}{\delta}}\right).$$

Remark 1 Since $\min\{2\hat{\beta}_1, 2\hat{\beta}_2\} \le 2\hat{\beta}_2$ and $\hat{\beta}_3 \le 2\hat{\beta}_2$, the cardinality factor in Theorem 14 is bounded by $2\hat{\beta}_2 \le 4\ln n + 4$. That is, with probability at least $1-\delta$, App-BM+ admits the following cardinality guarantee:

$$|B| \le (4\ln n + 4)\,|B_2^o| = \mathcal{O}(\ln n)\cdot|B_2^o|.$$

Notably, despite the non-submodularity of our objective, this cardinality ratio matches the order of the best possible guarantee for the submodular cover problem [13]. To the best of our knowledge, within the body of work on influence analysis, App-BM+ is the first practical approximation algorithm that provides strong performance guarantees for a non-submodular cover problem, i.e., minimizing the solution cardinality subject to a threshold constraint on a non-submodular objective. We believe that App-BM+, together with our theoretical analysis, is of independent interest and may be adapted to tackle other problems.

Remark 2 While this paper focuses on the IC model, the proposed techniques for both IMIN and BM problems naturally extend to another widely studied model, the linear threshold (LT) model [24]. Specifically, the objective $D(\cdot)$ has been proved to be submodular under the LT model [61], which allows us to optimize $D(\cdot)$ itself rather than resorting to surrogate bounds. This tightens the approximation guarantees, as the gap between the surrogates and the objective is no longer incurred. Moreover, our estimators can be adapted to the LT model by modifying the sampling procedure, as the diffusion process under LT also admits an equivalent live-edge characterization [24]. The optimization framework then applies directly, since it relies only on the statistical properties of the estimators.

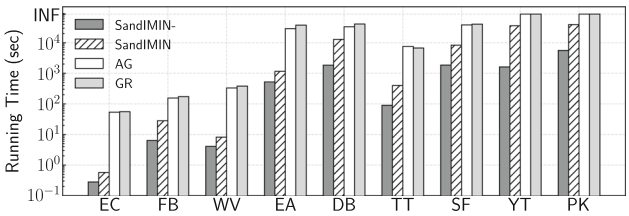


Fig. 3 Running time on all the datasets

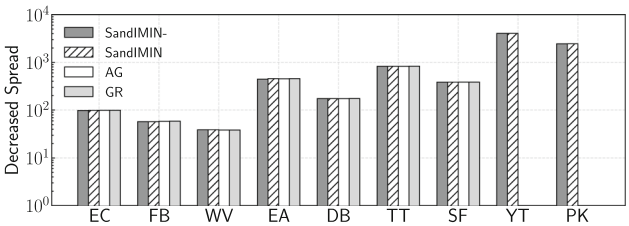


Fig. 4 Decreased spread on all the datasets

5 Experiments

In this section, we conduct extensive experiments to evaluate the performance of our proposed algorithms on both IMIN and BM problems. All the programs are implemented in C++ and performed on a PC with an Intel Xeon 2.10GHz CPU and 512GB memory. The source code is available at [1].

5.1 Experimental setup

Algorithms We evaluate the following algorithms on the IMIN problem. (1) AG and GR: the state-of-the-art algorithms proposed in [60]. (2) SandIMIN: the approximation algorithm for the IMIN problem proposed in this work. (3) SandIMIN-: a simplified version of SandIMIN in which the App-UB subroutine is not executed for improved efficiency.

For the BM problem, we evaluate the existing solution (1) Base [41], and our proposed three algorithms: (2) Heur-BM; (3) App-BM; (4) App-BM+.

Datasets We use 9 real-world datasets which are available on SNAP,[1] i.e., EmailCore, Facebook, Wiki-Vote, EmailAll, DBLP, Twitter, Stanford, Youtube and Pokec. The details of the datasets are reported in Table 2 and we provide a short name for each dataset.

Parameter settings Following the convention [22, 38, 43, 46, 47], we set the propagation probability $p(u, v)$ of each edge $\langle u, v\rangle$ as the inverse of v's in-degree in IC model. The seed nodes in S are randomly selected from the top 200 most influential nodes.

IMIN problem. By default, we set $\epsilon = 0.2$, $\beta = \gamma = 0.1$ and $\delta = 1/n$ for SandIMIN and SandIMIN-. For AG and GR, we set the number of generated realizations to 10,000, which is recommended in [60]. $|S|$ and k are 10 and 100 by default, respectively.

BM problem. We set $|S| = 10$ and $\eta = 0.2\hat{I}_G(S)$ by default, where $\hat{I}_G(S)$ is a $(0.1, 1/n)$-estimate of $\mathbb{E}[I_G(S)]$. We fix the number of trees generated by Base to 1,000, following the setting in [41]. In addition, we set $\delta = 1/n$ for Heur-BM, App-BM and App-BM+, $\theta_h = 10,000$ and $\gamma = 0.1$ for Heur-BM, and $\mu = 10^{-2}$ for App-BM. Notably, in the majority of datasets, $\frac{\mathbb{E}[I_G(S)]-|S|-\eta}{\mathbb{E}[I_G(S)]-|S|+\eta+2|V_s'|}$ can exceed 10^{-2}. Consequently, App-BM often fails to provide any approximation guarantees under the default settings, as indicated by Theorem 12.

Moreover, we estimate the influence spread of the seed set S by taking the average of its spreads over 10^5 Monte-Carlo simulations. For each parameter setting, we repeat each algorithm 10 times and report the average value. For those experiments that cannot finish within 24 h, we set them as **INF**.

5.2 Performance evaluation of the IMIN solutions

Efficiency evaluation In Fig. 3, we first evaluate the time cost on all the datasets with the default settings. As can be seen, SandIMIN and SandIMIN- cost less time than AG and GR on all the datasets and they can achieve up to two orders of magnitude speedup. In most cases, AG is more efficient than GR. Besides, AG and GR cannot complete on large datasets (i.e., Youtube and Pokec) in a reasonable time. The primary reasons are that our methods transform the original non-submodular maximization problem into the submodular maximization scenario. This allows us to avoid generating new samples after selecting each blocker. Furthermore, our heuristic algorithm for IMIN incurs almost no computation overhead. In addition, SandIMIN- is more efficient than SandIMIN on all the datasets, which is not surprising given that SandIMIN- does not execute App-UB. Moreover, as the dataset becomes large, the gap between the time cost of our solutions and that of AG and GR becomes smaller. This is because, to provide theoretical guarantees for our solutions, the number of samples generated will increase as the dataset becomes large, while AG and GR use a fixed number of realizations each round for graphs of any size and offer no approximation guarantee for the quality of the returned results.

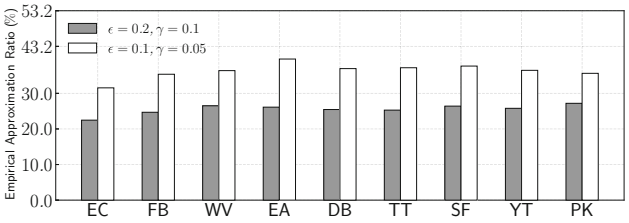


Fig. 5 Approximation ratio of SandIMIN on all the datasets.

Effectiveness evaluation In Fig. 4, we present the decreased spread on all the datasets with the default settings. The results indicate that our solutions always exhibit similar performance to AG and GR. Note that, on Youtube and Pokec, AG and GR cannot finish in a reasonable time. Therefore, the corresponding value is not shown in the figure. We also evaluate the approximation guarantees of SandIMIN. As presented in Theorem 5, the exact approximation ratio of SandIMIN is intractable to compute, because B_1^o and B_L^o are NP-hard to obtain. According to [56], $\frac{(1-\gamma)^2}{(1+\gamma)^2} \cdot (1 - 1/e - \epsilon) \cdot \frac{\hat{D}_s(B_U)}{\hat{D}_s^U(B_U)}$ is a computable lower bound of the approximation ratio for SandIMIN. Notably, no computable approximation ratio is provided for SandIMIN-, since it does not return B_U. The average lower bound of the approximation ratio of SandIMIN on all the datasets (averaged over $k = 10, 50$ and 100) is shown in Fig. 5. We report the results under two different settings of ϵ and γ. In particular, on all the datasets, the approximation ratio exceeds 20% with $\epsilon = 0.2, \gamma = 0.1$, and exceeds 30% with $\epsilon = 0.1, \gamma = 0.05$. As a data-dependent approximation guaranteed algorithm, SandIMIN performs well in terms of approximation, as the ratios closely approximate the value of $1 - 1/e - \epsilon$.

5.3 Performance evaluation of the BM solutions

Efficiency evaluation Figs. 6 and 7 report the running time of the BM approaches on all datasets when varying $|S|$ and $\eta/\hat{I}_G(S)$. We observe that Base achieves the highest efficiency, as it requires generating only a small number of samples. In contrast, Heur-BM must regenerate a certain portion of samples containing the selected blocks at each iteration; App-BM and App-BM+ demand a substantially larger number of samples to ensure the theoretical guarantees. In addition, App-BM always runs much slower than both Heur-BM and App-BM+, and is able to complete within a single day only on the relatively small datasets. This obser-

[1] http://snap.stanford.edu.

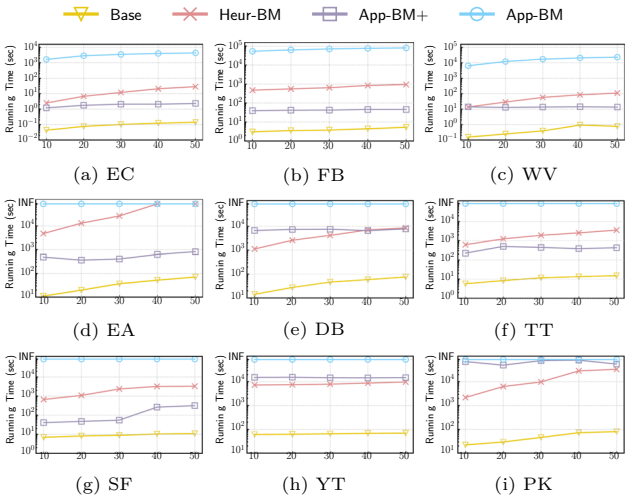


Fig. 6 Running time when varying $|S|$

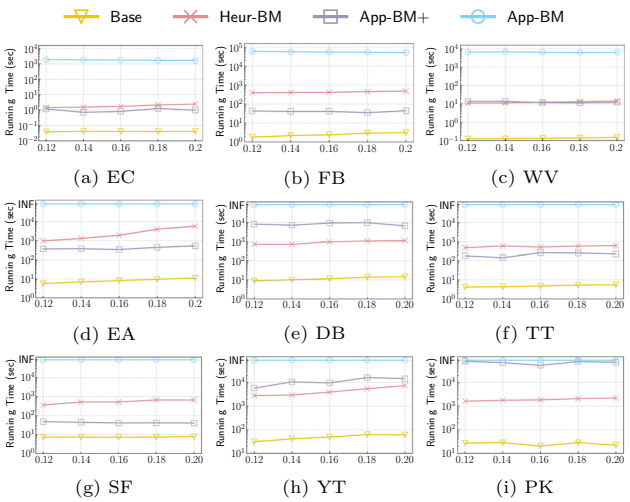


Fig. 7 Running time when varying $\eta/\hat{I}_G(S)$

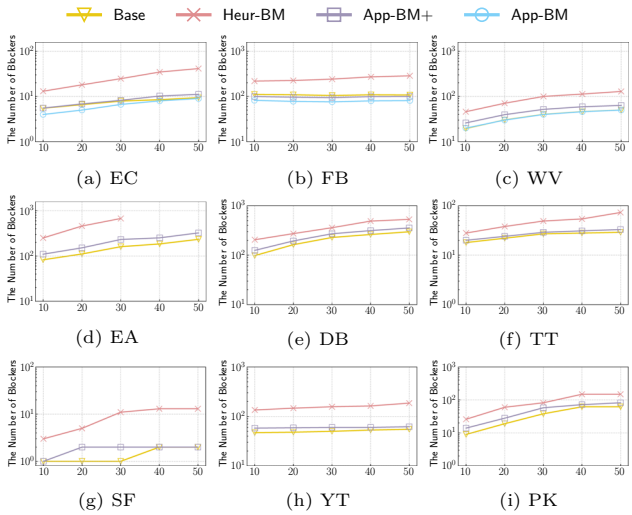


Fig. 8 The number of blockers when varying $|S|$

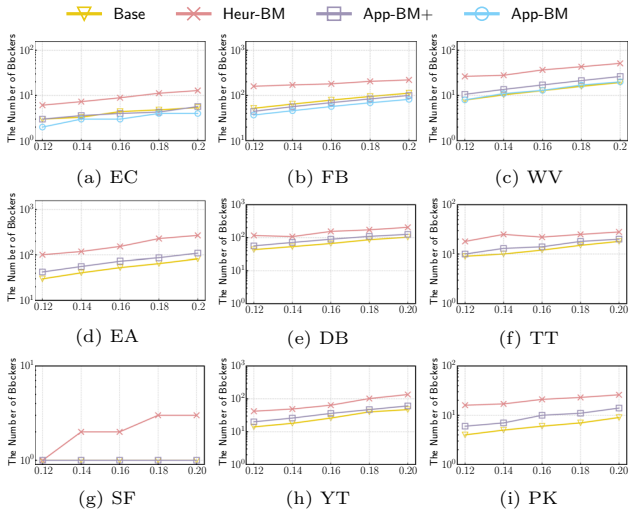


Fig. 9 The number of blockers when varying $\eta/\hat{I}_G(S)$

vation indicates that, although App-BM achieves a (slightly) tighter approximation ratio than App-BM+, its contribution is predominantly theoretical, while App-BM+ emerges as the more suitable option for practical deployment. Moreover, in most cases App-BM+ runs faster than Heur-BM, demonstrating the superiority of our trial-and-error sampling scheme. Only on the three large graphs (DBLP, Youtube, and Pokec) does Heur-BM outperform App-BM+. This is because Heur-BM adopts a fixed sampling size regardless of the dataset, whereas the stopping conditions of App-BM+ generally become harder to satisfy as the graph scales. As for the effect of the two parameters, it can be observed that the runtime of Heur-BM increases with larger values of $|S|$ or $\eta/\hat{I}_G(S)$. This is because an increase in these quantities results in a larger value of η, which necessitates selecting more blockers to satisfy the threshold condition. Each blocker selection further entails Heur-BM to regenerate multiple sample sets, thereby increasing the overall computational overhead. In contrast, App-BM+ shows relatively stable performance under varying values of $|S|$ and $\eta/\hat{I}_G(S)$, which highlights its robustness.

Effectiveness evaluation We then evaluate the effectiveness of our BM algorithms using two metrics: the cardinality of the returned blocker set and the *threshold satisfactory ratio*, defined as the ratio between the reduction in influence spread and the given threshold. A threshold satisfactory ratio above 1 indicates that the threshold condition is met. Figs. 8, 9, 10, 11 report the number of selected blockers and the threshold satisfaction ratio on all datasets when varying $|S|$ and $\eta/\hat{I}_G(S)$. It can be observed that, in almost all cases, the threshold satisfactory ratio achieved by Base is below 1; in EmailCore, the ratio even falls below 0.8. This inferior performance can be attributed to the use of a biased estimator in Base, which makes its stopping condition for selection

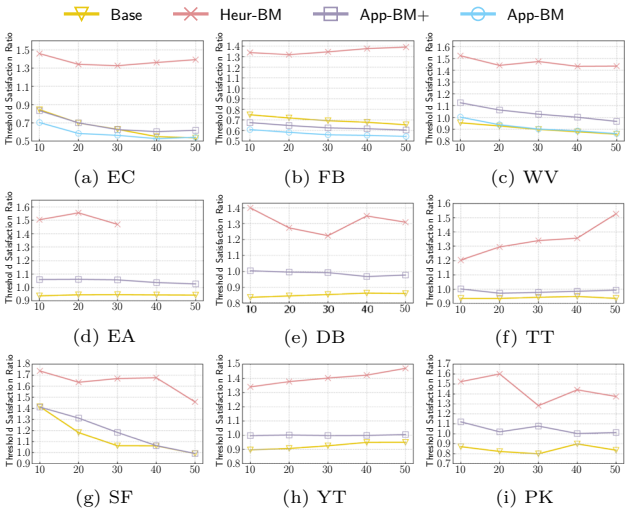


Fig. 10 Threshold satisfaction ratio when varying $|S|$

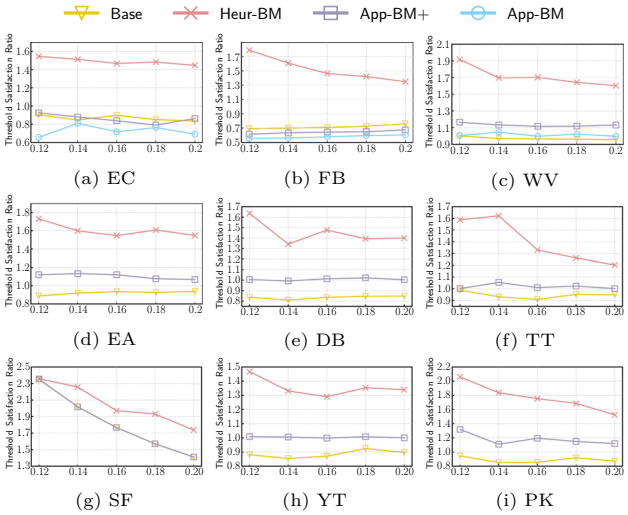


Fig. 11 Threshold satisfaction ratio when varying $\eta/\hat{I}_G(S)$

inaccurate. In particular, the estimator deviates considerably from the true objective value, causing the algorithm to terminate blocker selection well before the threshold condition is actually satisfied. This also explains why Base consistently returns a blocker set with the smallest cardinality. In addition, we find that Heur-BM often yields a threshold satisfactory ratio substantially greater than 1. One possible explanation is that, to theoretically guarantee the satisfaction of the threshold condition, the stopping criterion of Heur-BM in blocker selection tends to be overly conservative. The threshold satisfactory ratio offered by App-BM+ falls below 1 in a few datasets, indicating that App-BM+ may violate the threshold condition in certain scenarios. This outcome is not unexpected, as App-BM+ provides bicriteria approximation guarantees, which permit minor violations of the threshold condition. Furthermore, App-BM+ consistently requires fewer blockers than Heur-BM, owing to its $\mathcal{O}(\ln n)$ cardinality approximation guarantee. Notably, although Heur-BM provably admits no constant-factor bound on its cardinality (Lemma 11), in practice it does not fail drastically: over all datasets and settings, it returns less than one order of magnitude more blockers than App-BM+. To explain, the unbounded worst case in Lemma 11 arises only when a few nodes jointly protect a large region while each contributes a negligible marginal gain, so that the greedy overlooks them; such configurations are quite rare in practice.

Remark Theoretically, App-BM+ is favored over Heur-BM for its strong approximation guarantee, which certifies its solution quality in advance. Empirically, our experimental results show that on the small and medium datasets, App-BM+ generally dominates Heur-BM, returning fewer blockers while being faster than Heur-BM. On some large datasets, however, a trade-off emerges: App-BM+ still returns fewer blockers, but Heur-BM can run faster. In such cases, App-BM+ is preferable when a small blocking set is desired, whereas Heur-BM may be more suitable when runtime is the dominant concern.

6 Related work

The influence maximization (IM) problem, which aims to find a set of users with the largest influence spread, is a fundamental problem in graph analysis. Kempe et al. [24] first formulate IM problem and propose independent cascade (IC) as well as linear threshold (LT) models. In addition, they utilize a greedy algorithm that returns $(1-1/e-\epsilon)$-approximate solution. Afterwards, a large number of work [7–11, 17, 23, 39, 44, 45, 55] focuses on the development of heuristic algorithms to reduce the computation overhead. However, such solutions return the results without any approximation guarantees. As a milestone, Borgs et al. [5] propose the reverse influence sampling (RIS) technique for influence spread estimation. Subsequently, many RIS-based algorithms [18, 22, 38, 43, 46, 47] that ensure $(1-1/e-\epsilon)$-approximations with reduced computation overhead are proposed. To better capture real-world scenarios, numerous variants of IM have also been extensively investigated. For instance, [26, 31] incorporate temporal aspects, while [30, 53, 54] integrate location information into the IM setting. Wang et al. [50] study the IM problem with priority constraints. Moreover, the seed minimization (SM) problem, a classical dual of IM, has been widely studied in the literature [14, 16, 20, 21, 28, 32, 42, 62, 64]. The goal of SM is to identify a seed set of minimum cardinality that ensures at least η nodes are influenced in expectation, where η is a predefined threshold.

As the counterpart of IM, the influence minimization (IMIN) problem has also attracted great attention due to its wide applications (e.g., misinformation suppression) [3, 36]. Wang et al. [52] first formulate and study IMIN under the IC

model, proposing a greedy method based on Monte-Carlo simulations. More recently, Xie et al. [60] design a novel dominator tree–based approach to efficiently estimate the reduction in influence spread, on top of which two greedy methods are developed. However, we note that all existing IMIN solutions mentioned above are heuristic without any approximation guarantees. This paper fills this gap by proposing SandIMIN, an efficient and theoretically grounded algorithm for IMIN. Beyond the node-blocking strategy, several studies have explored alternative approaches to minimize the influence spread of a given seed set. Budak et al. [6] first propose to spread positive information to counterbalance the misinformation propagation. Under this strategy, the objective is shown to be monotonic and submodular. Leveraging these properties, Tong et al. [48, 49] later design the sampling based methods and present the algorithms that provide $(1 - 1/e - \epsilon)$-approximations. In [27] and [25], the edge-blocking strategy is investigated under the LT and IC model, respectively.

Furthermore, as a natural variant of IMIN, the BM problem has been formulated and investigated in the literature. In [40] and [41], Pham et al. study the BM problem under the LT and IC model, respectively. In both works, the authors propose scalable greedy algorithms equipped with tree-based estimation methods. However, their proposed solutions neither guarantee theoretical satisfaction of the threshold condition nor ensure that the cardinality (of the selected blocker set) is close to the optimal value. In this work, we alleviate these limitations by developing practical algorithms for BM with non-trivial approximation guarantees.

7 Conclusion

In this paper, we study the influence minimization and blocker minimization problems under the IC model. For the former, we propose the first approximation algorithm with superior efficiency and strong performance guarantees; for the latter, we design a scalable heuristic and two bicriteria approximation algorithms. Extensive experiments over 9 real-world datasets are conducted to verify the performance of our algorithms. A promising direction for future work is to study both problems under other diffusion models, such as time-aware models.

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